ON THE TRANSFER OF SHALIKA GERMS FOR G_2

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§ 0 INTRODUCTION

Motivation and rough outline. To apply the trace formula to the transfer of automorphic representations of two reductive groups G, H over a number field K, one must, among other things, have a comparison of local orbital integrals, for each local completion F of K. The comparison in the real and complex case was solved by Shelstad [S1]. In the p-adic case, such a local comparison - the smooth matching conjecture stated below - boils down, thanks to the work of R. Langlands and D. Shelstad ([LS1], [LS2]), to a certain matching of Shalika germs conjecture, also stated below. Shalika germs are mysteriously behaving functions associated to a semisimple element $s \in G(F)$ and a unipotent conjugacy class $\mathcal{O}$ of $M = Cent_{G(F)}(s)$, defined in a punctured compact-open neighborhood of s in the regular set [Sh]. They play a role in the analysis of p-adic orbital integrals on reductive groups roughly analogous to the role that the power functions play in the Taylor series of a function on the real line. By a well-known reduction known as Harish-Chandra descent, it usually suffices to assume that s is central and $M = G$. Langlands [L] introduced a geometric approach to the computation, or at least comparison, of Shalika germs. The comparison of Shalika germs associated to the 0-regular unipotent orbit was solved by Shelstad [S2], using results from [LS1], [LS2]. Langland's technique was further developed by T. Hales in his thesis [H1]. Hales concentrated on the comparison of 1-regular germs, matching the stable subregular germs in general and matching the κ-stable subregular germs for even rank unitary groups. He also worked out a number of the computations necessary to deal with the rank two case. This technique of Langlands goes by the name Igusa theory since it was Igusa [I] who developed a crucial asymptotic formula for an integral over fibers of a morphism $Y_C \to C$, where C is a curve and Y_C is a non-singular variety. This technique also relies on a formula for the coefficients of this expansion, involving principal value integrals over p-adic analytic manifolds, worked out in [I], [L] (see also [LS0]). In [L], [LS3], Langlands and Shelstad matched all the germs for groups of absolute type A_2 (see also [H3], where a descent argument is used instead of the direct comparison in [LS3]). In [H2], Hales worked out the comparison of germs for the forms of $B_2 = C_2$ using Igusa theory and a reduction/descent argument. In the present article, we verify the matching of Shalika germs conjecture mentioned above for the 1-, 2- and 3-regular germs for the exceptional group G_2, which, together with the result of [S2], proves the smooth matching theorem for G_2. In other words, our main result is the proof of the following

Smooth Matching Conjecture for G_2. *Let T_G denote an elliptic Cartan of $G = G_2$ which has an F-embedding into H and let κ denote a character of $H^1(Gal(\overline{F}/F), T_G)$ associated to a cuspidal endoscopic group $H = H(T_G, \kappa)$* [L] *(thus H is either $PGL(3)$ or $(SL(2) \times SL(2))/\{\pm 1\}$). Let T_H denote the image of an F-embedding of T_G into H. For each $f \in C_c^\infty(G(F))$ there is an $f^H \in C_c^\infty(H(F))$ for which*

$$\Delta_{G/H}^{T_G,\kappa}(\gamma_H, \gamma_\lambda)\Phi^{T_G,\kappa}(\gamma_G, f) = \Phi^{T_H,st}(\gamma_H, f^H),$$

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whenever γ_G is the image of a regular $\gamma_H \in T_H(F)$. (The notation is explained further below.)

Before sketching the ideas in the paper, we describe what is new or different from [H2], [H3], which this paper is comparable to. The notation will be explained later.

In §1, we carefully recall in detail the construction of the Igusa variety and the determination of local coordinates [L], [H1], [H2].

Section 2 gives the calculations for determining the action of the Galois group on the local coordinates on the Igusa variety for G_2 as in [H2].

Section 3 describes some new results: for any reductive group G, we produce specific patches generalizing those used in [H2] and show for G_2 that they cover the base Igusa variety Y'' (described below) at least up to codimension two. These specific patches are needed for doing certain local calculations. It is likely that these patches are regular under more general conditions, which would imply Y'' is regular in codimension one (conjectured in [H1]), but this is only proven in our special case. Anyway, by another method we show that, as a consequence of the theory of toroidal embeddings [K], Y'' is normal and Cohen-Macauley (see Lemma 3.7 below). This implies Hales' conjecture that Y'' is regular in codimension one in general.

In section 4 we compute the Igusa constants of all the base divisors on Y'' by an easy direct calculation and give an "endoscopic interpretation" of some of the calculations. We also compute the Igusa forms using local coordinates of Y_1.

In section 5, we first record some basic facts about the Igusa expansion over Y_C. Then we prove the geometric "no spurious divisors lemma" for G_2 which is basically an inequality $e(\theta,\beta) \leq 1$, where $e(\theta,\beta)$ is as in [L], which allows one to work on Y'' rather than its desingularization Y_C. Finally, we recall the statement of Hales' important "reduction lemma".

In sections 6, 7, and 8 we (a) compute coordinates on the 1-, 2- and 3-regular unipotent orbits (the latter two are not Richardson which is something new, though they are what I call "endoscopically Richardson"), (b) compute coordinates on the fiber $\phi^{-1}(\gamma_\lambda)^\Gamma$ (explained below), and (c) compute the Igusa form on the fiber and its explicit dependence on a tangent vector X of C. Not much in these sections is substantially different from [H2], [H3]. One new idea is that, roughly speaking, the subgroup generated by the one-parameter root-groups associated to roots "outside H" may be used to determine local coordinates for the "endoscopically Richardson" orbit $\mathscr{O} = \mathrm{Ind}_H^G(1)$.

The last section presents a conjecture which our method of proof seems to support. It seems to be the "ultimate" theorem which one might prove by this method without introducing any new ideas.

For the most part, the appendices collect explicit group-theoretic computations needed to justify a number of claims made in the main body of the paper. For example, we verify that the cuspidal endoscopic groups of G_2 are $PGL(3)$ and $(SL(2) \times SL(2))/\{\pm 1\}$.

Turning now to a brief discussion of the constructions we use, here is a rough sketch of Langlands' idea in its simplest form: Let $\gamma \in G(F)$ be an elliptic regular element and let $T_G = Cent_G(\gamma)^o$. The goal is to compute, or at least find some manageable expression for, the stable orbital integral $\int_{(T_G \backslash G)(F)} f(g\gamma g^{-1})dg$. (More generally, one can and does consider κ-stable orbital integrals.) The variety $T_G \backslash G$ may be identified, over the algebraic closure $\overline{F}$, with the variety of regular stars: let Ω denote the absolute Weyl group of T_G, B_G a Borel subgroup of G containing T_G, and S_0 denote the $G(\overline{F})$-orbit of the $|\Omega|$-tuple $(B_G^\omega)_{\omega \in \Omega}$, where G acts by conjugation on each coordinate. This orbit is called the variety of regular stars and since $\cap_{\omega \in \Omega} N_G(B_G^\omega) = T_G$, we have $S_0 \cong T_G \backslash G$ over $\overline{F}$. Let $\mathscr{B} \cong B_G \backslash G$ denote the Borel variety. Although $\mathscr{B}$ is complete, S_0 is not but conjugation induces a principal N_G-bundle structure on $S_0 \to \mathscr{B}$, where N_G denotes the unipotent radical of B_G, so S_0 may be roughly regarded as the base space $\mathscr{B}$ with fibers at each point isomorphic to

N_G. The Zariski closure of S_0 in $\mathscr{B}^{|\Omega|}$ is denoted S and is called the variety of stars. It is projective, complete, but contains singularities. It turns out that S_0 is Zariski open in S. Roughly speaking, the idea is to regard the above stable orbital integral as an integral over $S_0(F) \subset S(\overline{F})$. Let T'_G denote the subvariety of regular elements of T_G. We restrict γ to a regular curve C in $T'_G \cap \{1\}$ in a neighborhood of the identity and then construct a smooth variety Y_C with a fibering $\phi : Y_C \to C$, "the" smooth Igusa variety, whose fibers $\phi^{-1}(\gamma)$, $\gamma \in C,\ \gamma \neq 1$, are isomorphic to $T_G \setminus G$. In fact, the above stable orbital integral may be regarded now as a fiber integral parameterized by a local coordinate λ in a neighborhood of the singular point $1 \in C$. Taking the limit as λ tends to this singular point, one may then use techniques of Igusa [I], as developed by Langlands [L], to express the stable orbital integral as a sum involving principal value integrals over certain divisors contained in $\phi^{-1}(1) \subset Y_C$. By the homogeneity of the Shalika germs and the homogeneity of the coefficients in the Igusa expansion, only certain divisors can contribute to a given Shalika germ.

Here is a rough sketch of Igusa theory [L], [H1]: Assume G is F-quasisplit. Let C denote a 1-parameter subgroup of G in T_G, defined over F, containing the identity and no other singular point and whose tangent over $\overline{F}$ is not contained in any root hyperplane. Let $C' = C - \{1\}$. We sketch the construction of a non-singular algebraic variety Y_C, defined over F, for which

$$\begin{array}{ccccc} C' \times T_G \setminus G & \xrightarrow{\ i\ } & Y_C & \xrightarrow{\ \pi\ } & G \\ & & \downarrow \phi & & \\ & & C, & & \end{array}$$

with the following properties:
(a) if T_G and C are defined over F then so are Y_C, ϕ, π, and i;
(b) $\pi : Y_C \to G$ is proper and G-equivariant;
(c) $\phi : Y_C \to C$ is proper;
(d) $\pi(\phi^{-1}(1)) = V$, the unipotent variety of G;
(e) if $\mathscr{O} \subset G$ denotes a unipotent orbit and if $E = \pi^{-1}(u)$, for some $u \in \pi(Y_C) \cap \mathscr{O}$, then $\pi|_E : E \cap \pi^{-1}(\mathscr{O}) \to \mathscr{O}$ is faithfully flat [DG, Ch II, Proposition 3.1] and, in particular, surjective;
(f) $i(C' \times T_G \setminus G) \subset Y_C$ is open;
(g) $\pi \circ i(\gamma, g) = \gamma^g$ and $\phi \circ i(\gamma, g) = \gamma$.
(h) Y_C is $2p+1$-dimensional, where $p = |\Sigma^+|$ denotes the number of positive roots.
A divisor $E \subset Y_C$ for which $\pi(E) = \overline{\mathscr{O}}$ is called an $\mathscr{O}$**-divisor**. In general, although E is smooth, $\overline{\mathscr{O}}$ need not even be normal [KP, §0].
Step 1: Let X denote the Zariski closure of $T' \times T_G \setminus G$ in $G \times S$ and let Y denote the Zariski closure of $C' \times T_G \setminus G$ in X. Let $\chi : G \to T_G/\Omega$ denote the adjoint quotient [Slo], where Ω denotes the Weyl group of T_G. We will see below that X is covered by patches $X(B_+, B_-) \subset B_- \times S \times N_+$ and Y is covered by patches $Y(B_+, B_-) \times N_+ \hookrightarrow \chi^{-1}(C)N_- \times S \times N_+$, where N_-, N_+ denotes the unipotent radical of B_-, B_+, resp.. Here B_+, B_- are opposite Borels for which $T_+ = B_+ \cap B_-$ is defined over F. The completeness of S implies that the projections $Y \to G$ and $X \to G$ are proper.
Step 2: We construct below a resolution S_1 of S with a proper morphism $S_1 \to S$ and an injective morphism $S_0 \to S_1$. The image of S_0 in S_1 is denoted S''. Both S and S_1 are complete but S'' is not. Let X'', X_1 denote the Zariski closure of $T' \times T_G \setminus G$ in $G \times S''$, $G \times S_1$, respectively. Let Y'', Y_1 denote the Zariski closure of $C' \times T_G \setminus G$ in X'', X_1, respectively. Again, we have proper maps $Y_1 \to G$ and $X_1 \to G$. Though $Y'' \to G$ may not be proper, it is especially useful because of its relatively simple local coordinate relations. We can explicitly desingularize Y'' for $G = G_2$ by blowing up four times in succession. (The derivation of this is lengthy and not really needed for

the proof of our main result so is omitted.) Recall that the blow down morphisms are proper and defined over F. They are also G-equivariant by the universal property of blowing up [Har].
Step 3: Resolve the singularities of Y_1 contained in $Y_1 - Y''$ by using Mumford's semi-stable reduction theorem [K]. This yields a variety $\tilde{Y}_1$ and a blow down $\xi : \tilde{Y}_1 \to Y_1$ such that all the singularities of $\tilde{Y}_1$ map down to $Y_1 - Y''$ under ξ. Now perform an embedded resolution of singularities on $Y'' \hookrightarrow \tilde{Y}_1$ as in step 2. The resulting resolution of $\tilde{Y}_1$ is denoted Y_C. The blow down map $Y_C \to Y_1$ is proper, so π and ϕ are proper.

We remark that the properness of π and ϕ are needed to guarantee that the appropriate principal value integrals make sense. Properness implies that the principal value integral of the pull-back of the function f_κ is well-defined and that some of the hypotheses to Lemma 5.7 below hold.

Notation and detailed outline. Let $\gamma_\lambda = exp(\lambda X) \in C'(F) \subset T_G(F)$ be the image of $\gamma_{\lambda,H} \in T_H(F) \subset H(F)$ in the sense of [LS2]. After some general background, we will examine the 1-, 2-, and 3-regular germs for G_2 using the formula ([Sh], [LS2])

$$\begin{aligned}\Delta_{G/H}^{T_G,\kappa}(\gamma_{\lambda,H},\gamma_\lambda)\Phi^{T_G,\kappa}(\gamma_\lambda,f) &= \kappa(inv(\gamma_G^0,\gamma_\lambda))D_H(\gamma_{\lambda,H})^{-1}\int_{\phi^{-1}(\gamma_\lambda)^\Gamma}\pi^*(f)\kappa(t_*)\frac{|\omega|}{|\phi^*(d\lambda)|} \\ &= \sum_{\mathscr{O}}\mu_{\mathscr{O}}(f)\Gamma_{\mathscr{O}}^{T_G,\kappa}(\gamma_\lambda), \quad f \in C_c^\infty(G),\end{aligned} \tag{0.1}$$

where $\Delta_{G/H}^{T_G,\kappa}$ is the local transfer factor given in a neighborhood of the identity by

$$\Delta_{G/H}^{T_G,\kappa}(\gamma_{\lambda,H},\gamma_G) = \kappa(inv(\gamma_G^0,\gamma_G))D_G(\gamma_G)/D_H(\gamma_H)$$

as in [LS2], γ_G^0 being a fixed regular element of $T_G(F)$. The sum over $\mathscr{O}$ in (0.1) runs over all stable unipotent orbits $\mathscr{O}^{st}$ in $G(F)$, $\mathscr{O} \subset \mathscr{O}^{st}$ denoting an arbitrarily chosen $G(F)$-orbit within $\mathscr{O}^{st}$. Also, in (0.1), $\gamma_\lambda = \gamma_{\lambda,G} \in C \cap V_f \subset T_G$ has parameter λ, so that $\gamma_0 = 1$, where $V_f \subset G_{reg}$ is a punctured neighborhood of 1, t_* is a cocycle-valued function which will be defined following 0.6 below, and $\phi^{-1}(\gamma_\lambda)^\Gamma$ denotes the fixed points of the fiber under the action of $\Gamma := Gal(\overline{F}/F)$ on Y_C determined by T_G as described in §2 below. The measure $\frac{|\omega|}{|\phi^*(d\lambda)|}$ incorporates the factor $D_G(\gamma_{\lambda,G})$. If $\gamma'_G \in G(F)$ is stably conjugate to γ_G then $\Delta_{G/H}^{T_G,\kappa}(\gamma_H,\gamma'_G) = \kappa(inv(\gamma_G,\gamma'_G))\Delta_{G/H}^{T_G,\kappa}(\gamma_H,\gamma_G)$. We will normalize the measures and use the same notation as in [LS3]. In particular,

$$\Gamma_{\mathscr{O}}^{T,\kappa}(\gamma) = \Delta_{G/H}^{T,\kappa}(\gamma_H,\gamma)\sum_\delta \kappa(\delta)\Gamma_{\mathscr{O}}(\gamma^\delta),$$

if γ is an image of some γ_H. Here δ runs over classes in $H^1(\Gamma,T_G)$ associated to the distinct stable conjugacy classes $\{\gamma^\delta\}$ of γ. These germs $\Gamma_{\mathscr{O}}(\gamma)$ are slightly different from those in [Sh] since the germs here do not include the discriminant factor in their definition. By a result of Harish-Chandra, if $\mathscr{O}$ is r-regular then $\Gamma_{\mathscr{O}}$ satisfies

$$\Gamma_{\mathscr{O}}(exp(\lambda^2 X)) = |\lambda|^{-dim\ \mathscr{O}}\Gamma_{\mathscr{O}}(exp(X)),$$

where for any $u \in \mathscr{O}$, $dim\ \mathscr{O} = dim\ G_u \backslash G = dim\ G - rk(G) - 2r = 2|\Sigma_G^+| - 2r$ with G_u denoting the centralizer of u in G, $r = dim\ \mathscr{B}_u$, $\mathscr{B}$ denoting the Borel variety, $\mathscr{B}_u$ its subvariety fixed by the conjugation action of u and Σ_G^+ denoting the positive roots with respect to some Borel. Note that since

$$D_G(exp(\lambda X)) = \prod_{\sigma>0}(1-\sigma(exp(\lambda X))^{-1}) = \lambda^{|\Sigma_G^+|}\prod_{\sigma>0}\sigma(X) + \text{ higher order terms},$$

the normalized germ $\Gamma_{\mathscr{O}}^*(\gamma) := |D_G(\gamma)|\Gamma_{\mathscr{O}}(\gamma)$ satisfies $\Gamma_{\mathscr{O}}^*(exp(\lambda^2 X)) = |\lambda|^{2r}\Gamma_{\mathscr{O}}^*(exp(X))$.

The ultimate object of our desires is the following

Matching of Germs Conjecture [LS3]**.** *Let H be an endoscopic group associated to T,κ. Given an r-regular unipotent orbit $\mathcal{O}$ in $G(F)$ there is a finite collection of r-regular unipotent classes $\mathcal{O}_i$ in $H(F)$ and complex numbers a_i such that, for every Cartan T_H of H over F and every regular ray $\{\lambda X|\lambda\neq 0\}$ in the Lie algebra $\mathfrak{t}_H$, we have*

$$\Gamma_{\mathcal{O}}^{T,\kappa}(\gamma_\lambda)=\sum_i a_i\Gamma_{\mathcal{O}_i}^{T_H,st}(\gamma_\lambda),$$

for λ sufficiently small.

Our goal is to try to verify this for $G=G_2$ and $r=1,2,3$.

The expression in (0.1) may also be written in the form $1/D_H(\gamma_{\lambda,H})$ times

$$F(\lambda)=\int_{\phi^{-1}(\lambda)(F)} f_\kappa(y)|\omega_\lambda|, \tag{0.2}$$

where f_κ, as defined by (0.1) and (0.2), satisfies the conditions of being an Igusa integrand in the sense of [L]. Igusa's expansion [L] is an asymptotic expression of the form

$$F(\lambda)=\sum_{\theta,\beta}\theta(\lambda)|\lambda|^{\beta-1}\sum_{r=1}^{e(\theta,\beta)}(-\log_q|\lambda|)^{r-1}F_r(\theta,\beta,f_\kappa), \tag{0.3}$$

where θ is a unitary character of $F^\times$ and $\beta>0$ is a rational number for which $\mathcal{E}(\theta,\beta)\neq\emptyset$. Here $\mathcal{E}(\theta,\beta)$ denotes the set of irreducible components (called **branches**) in $\phi^{-1}(1)\subset Y_C$ such that $\theta^{a(E)}=\kappa_E$ and $\beta=b(E)/a(E)$, where $a(E)$, $b(E)$, κ_E, and $e(\theta,\beta)$ are as in [L]. As T. Hales has observed, since there are no logarithmic terms in the Shalika germ expansion, we have

Lemma 0.4. *If f_κ is as in (0.2) then $F_r(\theta,\beta,f_\kappa)=0$ for all $r>1$.*

By Howe, Harish-Chandra, and Rogawski [Ro], the trivial germ is a non-zero constant c_H. We describe how we will be investigating the other terms in (0.1) associated to the 1-regular, 2-regular, and 3-regular orbits $\mathcal{O}$. These will match with the trivial orbit in the germ expansion on H,

$$\Phi^{T_H,st}(\gamma_{\lambda,H},f^H)=\int_{\phi_H^{-1}(\gamma_{\lambda,H})^\Gamma}\pi^*(f^H)\frac{|\omega|}{|\phi^*(d\lambda)|}=\sum_{\mathcal{O}}\mu_{\mathcal{O}}(f^H)\Gamma_{\mathcal{O}}^{T_H,st}(\gamma_{\lambda,H}), \tag{0.5}$$

where $H=(SL(2)\times SL(2))/\{\pm1\}$ or $H=PGL(3)$, and where $f^H\in C_c^\infty(H)$ exists and $\Delta_{G/H}^{T_G,\kappa}$-matches with f. These terms are the $\beta=2,3,4$ terms in the Igusa expansions for G_2, $H_1=PGL(3)$ and $H_2=(SL(2)\times SL(2)))/\{\pm1\}$. It ultimately turns out that, in these expansions, we have $e(\theta,\beta)\leq 1$ (this is proved in §§4-5 below). Therefore, formula 0.6 below applies to our situation.

Putting together results of [L], [H2, §2], using the Shalika germ expansion, we have the following

Formula 0.6(a). *Assume that we have a Cartan T_G, a character κ of $H^1(\Gamma,T_G)$, and a $(\beta-1)$-regular stable unipotent class $\mathcal{O}^{st}$ for which there is only one F-rational irreducible branch E_β of $\{\lambda=0\}$ in Y_C with $\pi(E_\beta)=\overline{\mathcal{O}}^{st}$ and $\beta(E_\beta)=\beta$. Then we have*

$$D_H(\gamma_{H,\lambda})\sum_{\mathcal{O}\subset\mathcal{O}^{st}}\mu_{\mathcal{O}}(f)\Gamma_{\mathcal{O}}^{T_G,\kappa}(\gamma_\lambda)=\kappa(inv(\gamma_G^0,\gamma_\lambda))|\lambda|^{\beta-1}\sum_\theta\theta(\lambda)\int_{E_\beta(F)}\pi^*(f)\kappa_\theta(t_*)|\omega_{E_\beta}|,$$

for $\gamma_\lambda\in C\cap V_f$. Here $\kappa_\theta(t_*)$ is an extension of $\kappa(t_*)/\theta(\lambda)$ to E_β and $\overline{\mathcal{O}}^{st}$ denotes the F-Zariski closure of the stable orbit and t_* is a cocycle-valued function described more below, and the sum on the left runs over all unipotent $G(F)$-orbits $\mathcal{O}$ contained in the stable orbit $\mathcal{O}^{st}$.

0.6(b): Same assumptions. Then for each $\overline{\mathcal{O}}^{st}$ there is a measure $\omega_{\overline{\mathcal{O}}^{st}}$ and for each F-rational fiber $E_\beta(u) = E_\beta(F) \cap \pi^{-1}(u)$, $u \in \overline{\mathcal{O}}(\overline{F})$, there is a measure $\omega_{E_\beta(u)}$ such that

$$\int_{E_\beta(F)} \pi^*(f)\kappa_\theta(t_*)|\omega_{E_\beta}| = \int_{\overline{\mathcal{O}}^{st}} f \int_{E_\beta(u)} \kappa_\theta(t_*)|\omega_{\overline{\mathcal{O}}^{st}}||\omega_{E_\beta(u)}|,$$

for each θ occuring above.

This will follow from the results of section 5 below. By [H1, ch 6, §2], only the $\theta = 1$ term on the left side of formula 0.6(a) contributes a non-zero term to the sum. It turns out that either κ is quadratic (and $H = (SL(2) \times SL(2))/\{\pm 1\}$) or κ is a cubic character (and $H = PGL(3)$).

Now we turn to recalling some of the properties of t_* [L], [H1]. Recall $\Gamma = Gal(\overline{F}/F)$. The cocycle-valued function t_* is a map $Y_C \to Z^1(\Gamma, T_G(\overline{F}))$ given by $P \longmapsto (\sigma \longmapsto t_\sigma(P))$ which has the following properties:
(a) it is locally constant on the p-adic manifold $Y_C(F)$ [L];
(b) it factors through $Z^1(Gal(E/F), T_G(E))$, where E denotes the Galois closure of the splitting field of T_G;
(c) for fixed $\sigma \in \Gamma$, t_σ is a regular on the variety Y_C;
(d) for $y = i(\gamma_\lambda, g) \in Y_C$, such that for all $\sigma \in \Gamma$, $\sigma(g)g^{-1}$ belongs to $T_G(\overline{F})$, we have $\kappa(t_*(y)) = \kappa(\sigma(g)g^{-1})$, where $\sigma(g)g^{-1}$ represents the cocycle $\sigma \longmapsto \sigma(g)g^{-1}$.

After Y_C has been constructed, the idea is to determine the sets $\mathcal{E}(\theta, \beta)$ with $\beta = 2, 3, 4$. First, one determines the $\overline{F}$-branches in $\phi^{-1}(1) \subset Y_C(\overline{F})$ which map down to Y'' under $Y_C \to Y_1$ which have $\beta = 2, 3, 4$. These are called **fundamental divisors**. In fact, in the case of $G = G_2$, we can construct on Y'' itself divisors occurring in $\phi^{-1}(1) \subset Y''$. Such divisors will be called **base divisors**. Of course, a base divisor is fundamental. We sometimes identify a base divisor with its closure in Y_C. Then one determines which of these fundamental divisors has F-rational coordinates with respect to the Galois action determined by T_G. These branches will contribute to Igusa's expansion. Next, one must consider the $\overline{F}$-branches in $\phi^{-1}(1) \subset Y_C(\overline{F})$ which do not map down to Y'' under $Y_C \to Y_1$. These are called **spurious divisors** and are relatively complicated to deal with. It turns out that the computations in [H1, Ch III] show that there are no F-rational spurious divisors contributing to the subregular germ for G_2. For other reasons, the spurious divisors can be avoided altogether for the 2-regular and 3-regular germs. In fact, we have the following result, proven in §§4-5 below.

Lemma 0.8. *Let E be an F-rational base $\mathcal{O}$-divisor of the variety Y'' and let $\tilde{E}$ denote its strict transform in Y_C. The principal value integrals $\int_{E(F)} \pi^*(f)\kappa_1(t_*)|\omega_E|$ and $\int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa_1(\tilde{t}_*)|\omega_{\tilde{E}}|$ exist. Here κ_1 is as in formula 0.6(a), t_* is the cocycle function on Y'', and $\tilde{t}_*$ is the cocycle function on Y_C. Moreover $e(1, \beta_E) = 1$ and $F_1(1, \beta_E, f_\kappa)$ is given by $\kappa(inv(\gamma_G^0, \gamma_G))$ times*

$$\begin{aligned} PV \int_{E(F)} \pi^*(f)\kappa_1(t_*)|\omega_E| &= PV \int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa_1(\tilde{t}_*)|\omega_{\tilde{E}}| \\ &= \int_{\overline{\mathcal{O}}^{st}(F)} f \int_{E_u(F)} \kappa_1(t_*)|\omega_{E_u}||\omega_{\overline{\mathcal{O}}^{st}}|, \end{aligned}$$

where the measures are as in (0.6).

As a consequence of this, it will follow that spurious divisors do not contribute to the Igusa expansion of (0.1) or (0.2).

The proof that $e(\theta, \beta) \leq 1$ in the case of G_2 will now be sketched. From (0.1) we know that the germs associated to the $(\beta - 1)$-regular unipotent orbits $\mathcal{O}$ get "matched" with the terms in the Igusa expansion of homogeniety degree $\beta - 1$ in λ. Since both sides depend on X and λ we

might write such a comparison briefly as $\Gamma_\beta(\lambda,X,f) = I_\beta(\lambda,X,f)$, the left side denoting the sum of the normalized κ-stable germs of homogeniety degree $\beta-1$ and the right side denoting the corresponding term in the Igusa expansion (0.2), (0.3). If we replace X in both sides by cX then both sides should have the same degree of homogeniety in c as well. However, by computing explicitly the dependence of the term $I_\beta(\lambda,X,f)$ in the Igusa expansion on X we will see, using the case-by-case computation in §5, that $e(\theta,\beta) \geq 2$ implies that the two sides have different degrees of homogeniety in c, a contradiction.

It remains then to analyze the integrals over the fundamental divisors. We cannot compute these integrals but, in most cases, we can determine their dependence on λ and the tangent direction of C, and this is enough for our purposes. On the right-hand side of formula 0.6(a) we find what is essentially the factor

$$\Delta_0(X) = \lim_{\lambda\to 0} \frac{\Delta_{loc}(\gamma_{\lambda,G},\gamma_{\lambda,H})}{D_{G/H}(\gamma_{\lambda,G},\gamma_{\lambda,H})},$$

in the notation of [H3, §2]. The local translation invariance property in the hypotheses to Hales' reduction lemma 5.19 below holds for $\kappa(t_*)\Delta_0(X)$ as we will see (this is carried out in §§6-8 below, which we compute a quantity which is essentially $\Delta_0(X)$). Therefore, we leave the κ-stable $\mathscr{O}$-germ unchanged if we replace X by $X+Y$, where Y is sufficiently small and as in lemma 5.19. We may now apply lemma 5.18 to obtain the matching for functions whose support meets the stable orbit $\mathscr{O}^{st}(F)$ but no other unipotent orbit. (This is where [LS2] is implicitly used.) This descent argument, remarkably, allows us to avoid doing the difficult, more detailed calculations necessary for a direct determination of the dependence of $\Delta_0(X)\kappa(t_\sigma)$ on X.

As we've already mentioned, the 0-regular orbit has already been matched [S2]. The 1-regular κ-stable germ of G_2 matches with the 1-regular stable germ of $PGL(3)$ on cubic Cartans and with the 1-regular stable germ of $(SL(2)\times SL(2))/\{\pm 1\}$ on biquadratic Cartans. The stable germ associated to the trivial unipotent orbit of $(SL(2)\times SL(2))/\{\pm 1\}$ matches with the 2-regular κ-stable germ of G_2 on T, where T and κ determine $H_{T,\kappa} = (SL(2)\times SL(2))/\{\pm 1\}$. The stable germ associated to the trivial unipotent orbit of $PGL(3)$ matches with the 3-regular κ-stable germ of G_2 on T, where T and κ determine $H_{T,\kappa} = PGL(3)$. These matchings are in accordance with the fact that the $2-$regular and $3-$regular unipotent orbits of $G_2(\overline{F})$ are endoscopically induced from the trivial orbits of $A_1\times A_1$ and A_2, respectively [Spal, ch. III, table on page 221]. Finally, the matching for the trivial orbit follows from the work of Rogawski, Howe, and Harish-Chandra [Ro].

Though I've tried to make this paper self-contained (outside of the usual background in group theory [Bor] and algebraic geometry [M]), familiarity with the ideas in [L], [LS3], [H1], and [H3] is assumed.

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§ 1 Some Background

Let G denote a connected semi-simple quasi-split algebraic group defined over F and let T_G denote a maximal torus of G defined over F. In fact, we will assume that T_G is an elliptic Cartan but we will not make use of this immediately. We shall work over a separable closure $\overline{F}$ most of this section. A **root** is a character $\rho \in X^*(T)$ of a torus T such that there exists an isomorphism $\theta_\rho : \mathbb{G}_a \to U_\rho$, $U_\rho \subset G$ a closed subgroup, for which $t^{-1}\theta_\rho(\xi)t = \theta_\rho(\rho(t)\xi)$, for $t\in T$ and $\xi\in\mathbb{G}_a$. Here $\mathbb{G}_a$ denotes the 1-dimensional additive affine group. Let B_G denote a Borel containing T_G, Σ_G a set of roots of T_G, Σ_G^+ the set of positive roots with respect to B_G, and Δ the set of simple roots in

Σ_G^+. When $G = G_2$, we usually write $\Delta = \{\alpha, \beta\}$ and

$$\Sigma_G^+ = \{\alpha, \beta, \gamma = \alpha+\beta, \delta = 2\alpha+\beta, \varepsilon = 3\alpha+\beta, \zeta = 3\alpha+2\beta\}.$$

Fix a Borel $\underline{B}$ of G defined over F, a Cartan $\underline{T}$ defined over F contained in $\underline{B}$, and a positive Weyl chamber W_0 of T_G with respect to Σ_G^+.

Let Ω denote the absolute Weyl group of T_G. We may occasionally identify the roots of G with respect to T_G with those of G with respect to $\underline{T}$ by means of a fixed isomorphism $T_G \to \underline{T}$, and, thus, identify the absolute Weyl group of $\underline{T}$ with that of T_G. Representing $\omega \in \Omega$ by $w \in N_{G(\overline{F})}(T_G(\overline{F}))$, define

$$B_G(W) := w^{-1} B_G w = B_G^w,$$

for $W = \omega^{-1} W_0$. The Borels $\{B_G(W)) \mid W \in \mathscr{W}\}$ are precisely the Borel subgroups of G which contain T_G, where $\mathscr{W}$ denotes the set of Weyl chambers of T_G.

Recall that a **regular star** is a $|\Omega|$-tuple of Borels $(B(W))_{W\in\mathscr{W}}$ such that, for all $W \in \mathscr{W}$, $B(W) = B_G(W)^h = h^{-1}B_G(W)h$, for some $h \in G(\overline{F})$ (independent of W). The variety of regular stars is denoted by S_0. There is a natural bijection between $T_G(\overline{F}) \setminus G(\overline{F})$ and $S_0(\overline{F})$ given by $g \longmapsto (B(W)^g)_{W\in\mathscr{W}}$.

Let

$$X_0 := \{(g, (B(W))_{W\in\mathscr{W}}) \in G \times S_0 \mid g \in \cap_{W\in\mathscr{W}} B(W) \text{ is regular}\}, \tag{1.1}$$

so that we have an $\overline{F}$-isomorphism

$$\begin{aligned} T_G' \times T_G \setminus G &\to X_0 \\ (t, gT_G) &\longmapsto (g^{-1}tg, B_G(W)^g). \end{aligned} \tag{1.2}$$

Definition. *Let $\phi : X_0 \to T_G$ be the map which sends $(g,s) \in \mathscr{C}(T_G) \times S_0$ to the point in T_G which meets the conjugacy class of g, where $\mathscr{C}(T_G)$ denotes the union of the conjugates of T_G. Let $\pi : X_0 \to G$ denote the projection map onto the "G-coordinate".*

The map π restricts to an isomorphism onto the conjugacy class of γ,

$$\pi : \phi^{-1}(\gamma) \to \{g\gamma g^{-1} \mid g \in G/T_G\}.$$

Construction of Y_1. Let $C \subset T_G$ be a one-parameter subgroup of G containing $\{1\}$ but no other singular point. Let $C' := C - \{1\}$ and let $Y_0 := \phi^{-1}(C') \subset X_0$. The isomorphism (1.2) restricts to an $\overline{F}$-isomorphism

$$C' \times T_G \setminus G \to Y_0. \tag{1.4}$$

It can be shown that Y_0 is a regular algebraic subvariety of $G \times S$ [L, p. 480] but is not complete. Let Y denote the closure of Y_0 in $G \times S$. Y is singular in general.

Let $\mathscr{B} \cong B_G \setminus G$ denote the Borel variety. It will be helpful later to record here local coordinates for $\mathscr{B}$. First we recall the following well-known result [Bor, ch IV, Corollary 14.13]:

Lemma 1.5. *If $B, B', B'' \in \mathscr{B}$ and B', B'' are opposite to B then they are conjugate by an element of B.*

As a consequence of the Levi decomposition, it follows that, for B, B', B'' as in the lemma, B' and B'' are conjugate by an element of N, the unipotent radical of B. Let $\mathscr{B}(B) \subset \mathscr{B}$ denote the open subvariety of all Borels opposite to B.

Corollary 1.6. *For each $B \in \mathscr{B}$, $\mathscr{B}(B)$ is isomorphic as an affine variety to N and the coordinates on N (i.e., the matrix coefficients) serve as coordinates on $\mathscr{B}(B)$.*

A point $(B(W))_{W\in\mathscr{W}} \in \mathscr{B}^{|\Omega|}$ is called a **star** if for every $\omega\in\Omega$ and every simple root α of T_G in B_G there is an $h\in G(\overline{F})$ such that

$$B(\omega^{-1}W_0) = h^{-1}B_G h$$

and

$$B((s_\alpha\omega)^{-1}W_0) \subset h^{-1}P_{G,\alpha}h$$

where $P_{G,\alpha}$ is the smallest parabolic subgroup of G containing B_G and $B_G^{s_\alpha}$. Here s_α denotes the reflection associated to the simple root α of T_G in B_G. The variety of stars, denoted S, is closed in $\mathscr{B}^{|\Omega|}$, projective, but not in general regular as a variety. It is relatively easy to see that a regular star is a star. In fact, the variety of stars is the Zariski closure of S_0 in $\mathscr{B}^{|\Omega|}$. (One way to see this is by means of a more explicit realization of S_0 as a subset of S using coordinate patches (Lemma 1.2 below). That S_0 is open in its Zariski closure is a well-known result in group theory, e.g., [Sp3, ch.4, Lemma 4.3.1], [Bor, II.6, Prop. 6.7].)

We now define some open patches on S and regular coordinates on these patches. Fix a Borel B_+ of G and another Borel B_- which is opposite to B_+; let $T_+ := B_+\cap B_-$. Let

$$S(B_+) := \{(B(W))_{W\in\mathscr{W}} \in S \mid \forall W\in\mathscr{W},\ B(W) \text{ is opposite to } B_+\}$$

and let

$$S(B_+,B_-) := \{(B(W))_{W\in\mathscr{W}} \in S(B_+) \mid B(W_0) = B_-\}.$$

Let N_+ denote the unipotent radical of B_+. The map

$$((B(W))_{W\in\mathscr{W}}, n) \longmapsto (B(W)^n)_{W\in\mathscr{W}}$$

defines an isomorphism

$$S(B_+,B_-)\times N_+ \to S(B_+). \tag{1.7}$$

This is because for B_- and B_+ as above, the map $n\longmapsto B_-^n$, $n\in N_+$, defines an isomorphism from N_+ to the open subvariety $\mathscr{B}(B_+)\subset\mathscr{B}$ of all Borels opposite to B_+.

Following [L], we now describe local coordinates for $S(B_+,B_-)$.

We identify the absolute Weyl group of T_G with the absolute Weyl group of T_+. We identify the set of Weyl chambers of T_G with those of T_+. If (W,ρ) denotes a Weyl chamber and a root defining a wall of it (which we refer to below as a **wall-chamber pair** or often simply as a **pair**), we may find a $\omega\in\Omega$ for which $W=\omega^{-1}W_0$ and $\rho=\omega\alpha$, for some simple root α of T_+ in B_-; we sometimes write $\alpha(W,\rho)$ for the α obtained in this way.

Given a star $(B(W))_{W\in\mathscr{W}}\in S(B_+,B_-)$, to each pair (W,ρ) one defines the value at $(B(W))_{W\in\mathscr{W}}$ of the coordinate function $z(W,\rho)$ as follows. Let $ker(\rho)\subset X^*(T_G)\otimes\mathbb{R}$ denote the wall separating two adjacent Weyl chambers W, W' and let $h\in N_+(\overline{F})$ denote the unique element satisfying

$$B(W) = h^{-1}B_- h$$

and

$$B(W')\subset h^{-1}P_{-,\alpha}h$$

where $P_{-,\alpha}$ is the smallest parabolic subgroup of G containing B_- and $B_-^{s_\alpha}$, $\alpha=\alpha(W,\rho)$. Recall that

$$\mathbb{G}_a \cong U_{-\alpha} \cong P_{-,\alpha}/B_-,$$

via $z\longmapsto u=exp(X_{-\alpha})\longmapsto B_-^u$, which we may regard as an element of $P_{-,\alpha}/B_-\hookrightarrow G/B_-\cong\mathscr{B}$. Let $\mathbb{A}$ denote the affine line. There is a coordinate function $z(W,\rho)$,

$$z(W,\rho) : S(B_+,B_-)\to\mathbb{A},$$

such that

$$hB(W')h^{-1} = \theta_{-\alpha}(z(W,\rho))B_-\theta_{-\alpha}(-z(W,\rho)), \;\; \alpha = \alpha(W,\rho), \tag{1.8}$$

where X_α, $X_{-\alpha}$ are root vectors satisfying

$$[X_\alpha, X_{-\alpha}] = H_\alpha, \;\; \beta(H_\alpha) = 2\frac{(\beta,\alpha)}{(\alpha,\alpha)}, \;\; \theta_{-\alpha}(z) = \exp(zX_{-\alpha}) \in U_{-\alpha}.$$

The local coordinate $z(W_0,\alpha)$ on $S(B_+,B_-)$ is a local coordinate on a line of type α in the sense of [St]. This observation will be useful when dealing with the comparison of the subregular germ.

We have

(a) $z(W,\rho) = 0$ if and only if $B(W) = B(W')$ and

(b) the collection of the $z(W,\rho)$, for all pairs (W,ρ), uniquely determines the star $(B(W))_{W\in\mathscr{W}} \in S(B_+,B_-)$,

(c) the coordinate functions $z(W,\rho)$ are regular and generate the coordinate ring of $S(B_+,B_-)$.

The coordinates $z(W,\rho)$ are not algebraically independent. The coordinate relations they satisfy can be most naturally described using the notion of a gallery in the spherical building associated to the Weyl group Ω of T_G. We may identify the spherical building of T_G with that of T_+. Recall that a **gallery** is a sequence $W_0, W_1, \cdots, W_r$ of Weyl chambers such that W_i, W_{i+1} are distinct but share a wall $ker(\rho_{i+1})$. We always take W_0 to be the positive Weyl chamber of T_+ with respect to B_+ and define $\omega_i \in \Omega$ to be that element for which $W_i = \omega_i^{-1}W_0$. Let $\alpha_{i+1} := \omega_i\rho_{i+1}$, so $\alpha_{i+1} = \alpha(W_i,\rho_{i+1})$. If $z_i := z(W_{i-1},\rho_i)$ for $1 \leq i \leq r$, then

$$B(W_{i-1}) = h^{-1}B_-h, \quad B(W_i) = h^{-1}\theta_{-\alpha_i}(-z_i)B_-\theta_{-\alpha_i}(z_i)h, \tag{1.9}$$

for some $h \in N_+(\overline{F})$ possibly depending on i. We define n, n_{W_j}, for $1 \leq j \leq r$, by $B(W_j) = B_-^{n_{W_j}n}$, where $n, n_{W_j} \in N_+(\overline{F})$ and

$$n_{W_j} = \exp(z_jX_{-\alpha_j})\exp(z_{j-1}X_{-\alpha_{j-1}})\cdots\exp(z_1X_{-\alpha_1}). \tag{1.10}$$

In particular, if $W_r = W_0$ (so that the gallery is closed), then

$$\exp(z_rX_{-\alpha_r})\exp(z_{r-1}X_{-\alpha_{r-1}})\cdots\exp(z_1X_{-\alpha_1}) = 1. \tag{1.11}$$

In particular, if W, W' are adjacent and share a wall $ker(\rho)$ then $z(W,\rho) = -z(W',\rho)$.

We now state an observation of Langlands [L, pp.483-484]

Lemma 1.12. *TFAE:*
(a) A star $(B(W))_{W\in\mathscr{W}} \in S(B_+,B_-)$ is regular.
(b) $B(W) \neq B(W')$ for all adjacent W, W'.
(c) $z(W,\rho) \neq 0$ for all pairs (W,ρ).

We define coordinate patches on a resolution S_1 of S ([H1]).

Recall that all the short (resp. long) roots lie in the same orbit under the Weyl group. With this fact and the above lemma in mind, define

$$S'(B_+,B_-) := \{(B(W))_{W\in\mathscr{W}} \in S(B_+,B_-) \mid \forall\alpha\in\Delta, \exists\omega\in\Omega \text{ s.t. } z(\omega^{-1}W_0,\omega\alpha) \neq 0\}.$$

Suppose $(B(W))_{W\in\mathscr{W}} \in S(B_+,B_-)$ is not in $S'(B_+,B_-)$. Then $z(\omega^{-1}W_0,\omega\alpha) = 0$ for all $\alpha\in\Delta$ and $\omega\in\Omega$. By the lemma, this implies $B(W) = B(W')$ for all adjacent Weyl chambers. This is the diagonal in $\mathscr{B}^{|\Omega|}$ and the diagonal contains no regular stars.

Let $\mathbb{A}^\Delta$ denote the $|\Delta|$-fold Cartesian product of affine spaces and let T_+ act on $\mathbb{A}^\Delta$ by

$$t : (z(\alpha))_{\alpha\in\Delta} \longmapsto (\alpha(t)^{-1}z(\alpha))_{\alpha\in\Delta},$$

and on $S'(B_+,B_-)$ by

$$t : (z(W,\rho))_{(W,\rho)\text{ a pair}} \longmapsto (\alpha(t)z(W,\rho))_{(W,\rho)\text{ a pair}}, \quad \alpha = \alpha(W,\rho).$$

Let

$$S_1(B_+,B_-) := S'(B_+,B_-) \times_{T_+} \mathbb{A}^\Delta := (S'(B_+,B_-) \times \mathbb{A}^\Delta)/T_+. \tag{1.13}$$

Observe that there is a "blow-down" map

$$S_1(B_+,B_-) \to S(B_+,B_-), \tag{1.14}$$

given by $(s',z)T_+ \longmapsto s$, where $s' = (z_1(W,\rho))_{(W,\rho)\text{ a pair}}$, $z = (z(\alpha))_{\alpha\in\Delta}$ (note $z(\alpha)$ is only defined up to multiplication by $\alpha(t)^{-1}$), and

$$s = (z(W,\rho))_{(W,\rho)\text{ a pair}} = (z(\alpha(W,\rho))z_1(W,\rho))_{(W,\rho)\text{ a pair}}.$$

This is clearly well-defined (i.e., independent of the chosen representative $z = (z(\alpha))_{\alpha\in\Delta}$ of the T_+-orbit in $\mathbb{A}^\Delta$). The subscript 1 on $z_1(W,\rho)$ has been added to distinguish it from its image $z(W,\rho) := z(\alpha(W,\rho))z_1(W,\rho)$ in $S(B_+,B_-)$. Later we shall fix a canonical choice of a generic representative $(z(\alpha))_{\alpha\in\Delta}$ of the T_+-orbit in $\mathbb{A}^\Delta$: $z(\alpha)$ is chosen to be that multiple of $\alpha(t)^{-1}$ such that $z(\alpha) = \lambda/x(\alpha)$, where $x(\alpha)$ is as in (1.17) below (equivalently: $z(\alpha)$ is chosen to be that multiple which makes $w(\alpha) = 1$ for $\alpha > 0$ simple [H1, ch II, §2]).

Let $S'(B_+) \subset S(B_+)$ be the image of $S'(B_+,B_-) \times N_+$ in $S(B_+)$ under the map

$$((B(W))_{W\in\mathscr{W}}, n) \longmapsto (B(W)^n)_{W\in\mathscr{W}},$$

and let $S_1(B_+) \subset S'(B_+) \times_{T_+} \mathbb{A}^\Delta$ be the image of $S_1(B_+,B_-) \times N_+$ under

$$((B(W))_{W\in\mathscr{W}}, (z(\alpha))_{\alpha\in\Delta}, n) \longmapsto ((B(W)^n)_{W\in\mathscr{W}}, (z(\alpha))_{\alpha\in\Delta}).$$

Therefore,

$$S'(B_+) \cong S'(B_+,B_-) \times N_+, \tag{1.15a}$$

and

$$S_1(B_+) \cong S_1(B_+,B_-) \times N_+. \tag{1.15b}$$

The apparent dependence of $S_1(B_+)$ on B_- is not serious: by [L, Lemma 3.6], if we replace B_- by another Borel B'_- opposite to B_+ then the corresponding variety is canonically isomorphic to $S_1(B_+)$. Let B'_+ be another Borel and let B_- be a Borel opposite to both B_+ and B'_+. There is a birational morphism from $S_1(B_+)$ to $S_1(B'_+)$ as defined in [L]. Using this birational map to paste $S_1(B_+)$ and $S_1(B'_+)$ together, we obtain a variety S_1.

Observe that the "obvious" map $S_0(B_+) \to S_1(B_+)$ extends to a morphism $S_0 \to S_1$. This map yields a map $X_0 \to G \times S_1$.

Definition 1.16. *Let X be the Zariski closure of X_0 in $G \times S$. Let X_1 be the closure of X_0 in $G \times S_1$ and let Y_1 be the closure of Y_0 in X_1. Let $X_1(B_+)$ denote the intersection of X_1 with $G \times S_1(B_+)$ and similarly for $Y_1(B_+)$. Similarly, we define X'', Y'', $X''(B_+)$, $Y''(B_+)$, replacing $G \times S_1(B_+)$ by $G \times S''$ or $G \times S''(B_+)$ in the definition of X_1 and Y_1, where S'' is the image of S_0 in Y_1.*

In particular, the first coordinate of a point $p \in Y''(B_+,B_-) \subset G \times S''$ is an element b of $\cap_W B_-^{n_W} \subset B_-$, which we may write as an ordered product

$$b = t_0 \prod_{\rho\in\Sigma_G^+} \exp(x(\rho)X_\rho), \tag{1.17}$$

where Σ_G^+ denotes the set of roots of T_+ which are positive with respect to the Borel B_-, on which we fix an ordering once and for all.

Coordinates on Y''. Consider a point $x \in X_1 \subset G \times S_1$. On the patch of X_1 meeting $G \times S_1(B_+,B_-) \times N_+ \cong G \times S_1(B_+)$, we write x as $(b,(B(W))_{W\in\mathscr{W}},(z(\alpha))_{\alpha\in\Delta},n)$, where $b \in \cap_{W\in\mathscr{W}} B(W) \subset B_-$ and $B(W_0) = B_-$ (the condition that $b \in \cap_{W\in\mathscr{W}} B(W)$ is inherited from the definition of X_0).

Other coordinates on $X''(B_+)$ may be defined in terms of a point $(t^g,(B(W)^g)_{W\in\mathscr{W}})$ in $G \times S(B_+,B_-)$, where $t \in T_G(\overline{F})$ is regular. We may write

$$g = g_0 t_+ n_- n_+,$$

where $T_G^{g_0} = T_+$, $t_+ \in T_+(\overline{F})$, $n_- \in N_-(\overline{F})$, and $n_+ \in N_+(\overline{F})$. Let $t^{g_0} = t'_+$, for some $t'_+ \in T_+(\overline{F})$, and write $t^g = (t'_+ n'_-)^{n_+}$, for some $n'_- \in N_-(\overline{F})$. Therefore,

$$(t^g,(B(W)^g)_{W\in\mathscr{W}}) = (t'_+ n'_-, B_-^{n_W})^{n_+},$$

where $n_W \in N_+(\overline{F})$. We write

$$n'_- = \prod_{\rho\in\Sigma_G^+} \exp(x(\rho)X_\rho),$$

$$n_+ = \prod_{\rho\in\Sigma_G^+} \exp(v(\rho)X_{-\rho}),$$

where, by (1.10),

$$n_W = \exp(z(W_r,\rho_r)X_{-\alpha_r}) \exp(z(W_{r-1},\rho_{r-1})X_{-\alpha_{r-1}}) \cdots \exp(z(W_1,\rho_1)X_{-\alpha_1}), \tag{1.18}$$

where $W = \omega^{-1}W_0$, $\omega = s_{\alpha_r}\cdots s_{\alpha_1}$, and W_0, ..., W_r being chambers in a path from the positive Weyl chamber W_0 to $W_r = W$. Recall we defined $z_1(W,\rho)$, for any pair (W,ρ), by

$$z(W,\rho) = z(\alpha(W,\rho))z_1(W,\rho). \tag{1.19}$$

Define the coordinate function $w(\rho)$, $\rho > 0$, by $w(\alpha) = 1$, if $\alpha > 0$ is simple, and otherwise by

$$(t'_+)^{-1}(t'_+)^{n_-} = \prod_{\rho\in\Sigma_G^+} \exp\Big(\frac{w(\rho)\lambda}{\prod_{\alpha\in\Delta} z(\alpha)^{m(\alpha)}} X_\rho\Big), \tag{1.20}$$

where $\rho = \sum_{\alpha\in\Delta} m(\alpha)\alpha$. This describes a point in the image

$$C' \times G/T_G \hookrightarrow Y_1(B_+,B_-) \subset X_1(B_+,B_-) \subset G \times S_1(B_+,B_-),$$

where $C' = C \cap T'_G = C - \{1\}$.

Coordinates on Y_1. We begin with the following

Lemma 1.21. *(*[H1, ch II, Lemma 3.3]*) If $t \in T_+$ and*

$$n = \prod_{\rho>0} \theta_\rho(n_\rho) \in N_+ \quad (\textit{ordered product}),$$

then

$$t^n = t \prod_{\rho>0} \theta_\rho(y(\rho)),$$

where

$$y(\rho) = \sum_{\rho_1,\ldots,\rho_m} c_{\rho_1,\ldots,\rho_m}(t) n_{\rho_1} \ldots n_{\rho_m},$$

the sum ranging over all roots $\rho_i > 0$ such that $\rho = \rho_1 + ... + \rho_m$ and

$$c_{\rho_1,...,\rho_m}(t) = \sum_{\sigma_1,\sigma_2} (1-\sigma_1(t)^{-1})\sigma_2(t),$$

where the sum over σ_1, σ_2 such that σ_1 is a root and σ_2 is a sum of roots. Moreover, $c_\rho(t) = (1-\rho(t)^{-1})$.

We want to apply this lemma to obtain coordinates of a point

$$(t^g, (B(W))_{W\in\mathscr{W}}, (z(\alpha))_{\alpha\in\Delta})^n \in \chi^{-1}(C) \times S_1(B_+, B_-) \times N_+.$$

For the case $G = G_2$, we give coordinate relations between the coordinates in (1.20) and the n_ρ's and the $z(W,\alpha)$'s (see [H1, chap II]):

(1.22)
$$x(\alpha) = (1-\alpha^{-1})n_\alpha = (1-\alpha^{-1})/z(W_0,\alpha),$$
$$x(\beta) = (1-\beta^{-1})n_\beta = (1-\beta^{-1})/z(W_0,\beta),$$
$$x(\gamma) = (1-\gamma^{-1})n_\gamma + c_1 \cdot \frac{(1-\alpha^{-1})}{z(W_0,\alpha)z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)} + \text{ higher order terms in } \lambda,$$
$$x(\delta) = (1-\delta^{-1})n_\delta + c_2 \cdot \frac{(1-\alpha^{-1})}{z(W_0,\alpha)z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)} + \text{ higher order terms in } \lambda,$$
$$x(\varepsilon) = (1-\varepsilon^{-1})n_\varepsilon + c_3 \cdot \frac{(1-\alpha^{-1})n_\delta}{z(W_0,\alpha)} + \text{ higher order terms in } \lambda,$$
$$x(\zeta) = (1-\zeta^{-1})n_\zeta + c_4 \cdot \frac{(1-\alpha^{-1})n_\gamma^2}{z(W_0,\alpha)} + c_5 \cdot \frac{(1-\alpha^{-1})n_\delta}{z(W_0,\alpha)z(W_0,\beta)} +$$
$$+ c_6 \cdot \frac{(1-\beta^{-1})n_\varepsilon}{z(W_0,\beta)} + c_7 \cdot \frac{(1-\gamma^{-1})n_\delta}{z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)} + \text{ higher order terms in } \lambda,$$
$$w(\gamma) = \frac{x(\gamma)z(\beta)z(\alpha)}{\lambda}$$
$$= (\gamma(X)n_\gamma + c_1\frac{\alpha(X)}{z(W_0,\alpha)z(W_0,\beta)})\alpha(X)^{-1}\beta(X)^{-1}z(W_0,\alpha)z(W_0,\beta) +$$
$$+ \text{ higher order terms in } \lambda,$$
$$w(\delta) = \frac{x(\delta)z(\beta)z(\alpha)^2}{\lambda}$$
$$= (\delta(X)n_\delta + c_2\frac{\alpha(X)}{z(W_0,\alpha)z(W_0,\beta)z(\sigma_\beta^{-1}W_0,\alpha)})\alpha(X)^{-2}\beta(X)^{-1}z(W_0,\alpha)^2z(W_0,\beta) +$$
$$+ \text{ higher order terms in } \lambda,$$
$$w(\varepsilon) = \frac{x(\varepsilon)z(\beta)z(\alpha)^3}{\lambda}$$
$$= (\varepsilon(X)n_\varepsilon + c_3\frac{\alpha(X)n_\delta}{z(W_0,\alpha)})\alpha(X)^{-3}\beta(X)^{-1}z(W_0,\alpha)^3z(W_0,\beta) +$$
$$+ \text{ higher order terms in } \lambda,$$

$$w(\zeta) = \frac{x(\zeta)z(\beta)^2 z(\alpha)^3}{\lambda}$$
$$= (\zeta(X)n_\zeta + c_4\frac{\alpha(X)n_\gamma^2}{z(W_0,\alpha)} + c_5\frac{\alpha(X)n_\delta}{z(W_0,\alpha)z(W_0,\beta)} + c_6\frac{\beta(X)n_\varepsilon}{z(W_0,\beta)} +$$
$$+ \frac{\gamma(X)n_\zeta}{z(\sigma_\beta^{-1}W_0,\alpha)z(W_0,\beta)})\alpha(X)^{-3}\beta(X)^{-2}z(W_0,\alpha)^3 z(W_0,\beta)^2 +$$
$$+ \text{ higher order terms in } \lambda,$$

Because of (1.22), on $Y_1(B_+,B_-)$, we may use the algebraically independent coordinates

$$z(W_0,\alpha),\ z(W_0,\beta),\ z(\sigma_\beta^{-1}W_0,\alpha),\ n_\gamma,\ n_\delta,\ n_\varepsilon,\ n_\zeta,$$

in place of the coordinates

$$z(\alpha),\ z(\beta),\ x(\alpha),\ x(\beta),\ x(\gamma),\ x(\delta),\ x(\varepsilon),\ x(\zeta),\ w(\gamma),\ w(\delta),\ w(\varepsilon),\ w(\zeta),$$

on $Y''(B_+,B_-)$ appearing in (1.17) and (1.20).

§ 2 Galois action on the coordinates: explicit computations for G_2

Let G denote a connected semisimple F-quasi-split group with no outer automorphisms.

If T is a Cartan of G defined over F then $\Gamma = Gal(\overline{F}/F)$ preserves T and the set Σ of roots of G with respect to T, $\sigma \in \Gamma$ sending U_ρ to $U_{\sigma\rho}$, where U_ρ is a root group as in §1. Generally speaking, Γ will not preserve the set Δ of positive simple roots with respect to a Borel B containing T, unless B is defined over F. The character lattice $X := X^*(T) = X^*(T/\overline{F})$ is a $\mathbb{Z}[\Gamma]$-module and the action of Γ on $X_{\mathbb{R}} := X^*(T)\otimes\mathbb{R}$ preserves the set of root hyperplanes. Consequently, Γ acts on the set of Weyl chambers in $X_{\mathbb{R}}$ and preserves the positive Weyl chamber if B is defined over F. The action of $\sigma \in \Gamma$ on the set $\mathscr{W}$ of Weyl chambers in $X^*(T_G)\otimes\mathbb{R}$ will be denoted σ_T.

As in section 1, we fix a Borel $\underline{B}$ of G defined over F, a Cartan $\underline{T}$ defined over F contained in $\underline{B}$, and a positive Weyl chamber W_0 with respect to the linear order on Σ_G^+ which makes all the roots occuring in $\underline{B}$ positive. Let T_G denote our fixed Cartan of G defined over F and let B_G denote a Borel containing T_G. Note that, for any $\sigma \in \Gamma$ both B_G and B_G^σ are Borels containing T_G, since T_G is defined over F. The Galois action on $\underline{T}$ induces a homomorphism

$$\rho : \Gamma \to Aut(X,\Delta,X^\vee,\Delta^\vee), \tag{2.1a}$$

where $X = X^*(\underline{T})$ denotes the root lattice, $X^\vee = X_*(\underline{T})$ denotes the cocharacter lattice, $\Delta = \Delta(\underline{T},\underline{B})$ the simple roots of $\underline{T}$ with respect to $\underline{B}$, and $\Delta^\vee = \Delta^\vee(\underline{T},\underline{B})$ denotes the simple coroots. This induces ([Sp2]) a homomorphism

$$\tilde{\rho} : \Gamma \to Aut(G,B,T,\{X_\alpha\}_{\alpha\in\Delta}\}), \tag{2.1b}$$

where X_α denotes a root vector associated to α in the Lie algebra of G.

The Borels $\{B_G(W)) \mid W \in \mathscr{W}\}$ are precisely the Borel subgroups of G which contain T_G and, hence, are preserved by the Galois action since T_G is defined over F. Define a map $\sigma_T : \mathscr{W} \to \mathscr{W}$ by

$$\sigma B_G(W) = B_G(\sigma_T W),$$

for $\sigma \in \Gamma$. This defines a continuous homomorphism $\Gamma \to \Omega$, $\sigma \longmapsto \sigma_T$, which factors through $Gal(\overline{F}/E)$ where E/F denotes the splitting field of T_G. We may write $B_G = \underline{B}^g$, for some $g \in G(\overline{F})$ such that $g^{-1}\cdot\sigma g \in T_G(\overline{F})$ for all $\sigma \in \Gamma$. This action σ_T on the chambers $W \in \mathscr{W}$ induces an action, also denoted σ_T, on the positive roots Σ_G^+ and hence on the walls $ker(\rho)$ of W to the walls $ker(\sigma_T(\rho))$ of $\sigma_T W$.

We define an action of Γ on the set of regular stars, denoted S_0, by

$$\sigma(B(W)) := B_G(W)^{\sigma h} = \sigma B(\sigma_T^{-1}W), \tag{2.2}$$

where $B(W) = B_G(W)^h$. The set of F-rational points of $S_0(\overline{F})$ under this Γ-action is the set of regular stars $(B(W))_{W\in\mathscr{W}}$ such that $B(W) = B_G(W)^h$, with $\sigma(h)h^{-1} \in T_G(\overline{F})$ for all $\sigma \in \Gamma$.

For $\gamma \in T_G(F)$, let

$$\phi_\gamma^{-1}(F) := \phi^{-1}(\gamma)^\Gamma = \{x \in X^0(F) \mid \phi(x) = \gamma\},$$

where ϕ is defined following (1.2) above. In fact, the projection map π of §1.1 yields a set-theoretic bijection

$$\phi_\gamma^{-1}(F) \to \{g^{-1}\gamma g \mid g \in (T_G \setminus G)(F)\} \cong (T_G \setminus G)(F), \tag{2.3}$$

onto the stable conjugacy class of γ.

The action of $\sigma \in \Gamma$ on a point $((B_-^{n_W n})_{W\in\mathscr{W}}) \in S(B_+)$, for some $n, n_W \in N_+$, which we identify with the point $((B(W))_{W\in\mathscr{W}}, n) \in S(B_+, B_-) \times N_+$, is given by

$$\sigma : S(B_+, B_-) \times N_+ \to S(B_+, B_-) \times N_+,$$

$$((B(W))_{W\in\mathscr{W}}, n) \longmapsto ((B'(W))_{W\in\mathscr{W}}, n'), \tag{2.4a}$$

where $\sigma B_- = B_-^{c_\sigma}$, $c_\sigma \in N_{Aut\ G}(T_+)(\overline{F}) = N_G(T_+)(\overline{F})$, and

$$B'(W)^{n'} = \sigma(B(\sigma_T^{-1}(W))^n) = (\sigma B_-)^{\sigma(n_{\sigma_T^{-1}(W)}n)}, \tag{2.4b}$$

and

$$n' = c_\sigma \sigma(n_{\sigma_T^{-1}(W_0)}n), \;\; B'(W_0) = B_- \; . \tag{2.4c}$$

How does the action of Γ on $Y_1(B_+)$ affect the point determined by $(b, (B_-^{n_W})_{W\in\mathscr{W}})^n$? Here $b \in B_-$, $n_W, n \in N_+(\overline{F})$. Let $\alpha(W,\rho) \in \Delta$ denote the simple root associated to a pair (W,ρ) as above. To be consistent with (2.4a), define an action of $\sigma \in \Gamma$ on Δ, denoted by σ_T^*, as follows:

$$(\sigma_T^*)^{-1}(\alpha(W,\rho)) = \alpha(\sigma_T^{-1}(W), \sigma_T^{-1}(\rho)). \tag{2.5}$$

The action of Γ on $S_1(B_+) \cong (S'(B_+, B_-) \times_{T_+} \mathbb{A}^\Delta) \times N_+$, is given by

$$\sigma : ((B(W))_{W\in\mathscr{W}}, (z(\alpha))_{\alpha\in\Delta}, n) \longmapsto ((B'(W))_{W\in\mathscr{W}}, (z'(\alpha))_{\alpha\in\Delta}, n'),$$

where $B'(W)$ and n' are as above, and

$$z'(\alpha) = \sigma(z((\sigma_T^*)^{-1}(\alpha))).$$

By the formulas for the Galois action on $S(B_+)$, we have

$$\sigma : (b, (B_-^{n_W})_{W\in\mathscr{W}})^n \longmapsto (\sigma b, (\sigma B_-^{\sigma(n_{\sigma_T^{-1}(W)})})_{W\in\mathscr{W}})^{\sigma(n_{\sigma_T^{-1}(W_0)}n)}. \tag{2.6}$$

Here b is as in (1.17) and

$$n_W = \exp(z(W_r, \rho_r)X_{-\alpha_r}) \exp(z(W_{r-1}, \rho_{r-1})X_{-\alpha_{r-1}}) \cdots \exp(z(W_1, \rho_1)X_{-\alpha_1}),$$

where $W_0, ..., W_r = W$ is a gallery as in (1.9-10). How does the action of Γ on $S(B_+)$ affect the coordinates $z(W,\rho)$? A point of $S(B_+, B_-)$ is determined by $(z(W,\rho))_{(W,\rho)\ a\ pair}$, so by (2.2) and (2.6),

$$\sigma : (z(W,\rho))_{(W,\rho)\ a\ pair} \longmapsto (z'(W,\rho))_{(W,\rho)\ a\ pair},$$

where

$$z'(W,\rho) = \sigma(z(\sigma_T^{-1}(W), \sigma_T^{-1}(\rho)). \tag{2.7}$$

Suppose now that we are on a patch $S(B_+, B_-)$ with $T_+ = B_+ \cap B_-$ defined over F. There is an $c_\sigma \in N_{Aut\ G}(T_+)(\overline{F}) = N_G(T_+)(\overline{F})$ such that $\sigma B_- = c_\sigma B_- c_\sigma^{-1}$. We may then write

$$\sigma : (b, (B_-^{n_W})_{W \in \mathscr{W}})^n \longmapsto (b', (B_-^{n'_W})_{W \in \mathscr{W}})^{n'}, \tag{2.8}$$

where

$$b' = \sigma(n_{\sigma_T^{-1}(W_0)}) c_\sigma b c_\sigma \sigma(n_{\sigma_T^{-1}(W_0)}^{-1})^{-1},$$

$$n'_W = c_\sigma \sigma(n_{\sigma_T^{-1}(W)} n_{\sigma_T^{-1}(W_0)}^{-1}) c_\sigma^{-1},$$

$$n' = c_\sigma \sigma(n_{\sigma_T^{-1}(W_0)} n).$$

The action of c_σ by conjugation commutes with the action of σ_T by permutation of the Borels in a star. If B_- is defined over F then the W_0-component of $(B_-^{n'_W})_{W \in \mathscr{W}}$ is $\sigma B_- = B_-$. In particular, if the point in $Y_1(B_+)$ determined by $(b, (B_-^{n_W})_{W \in \mathscr{W}}, (z(\alpha))_{\alpha \in \Delta}, n_+)$ has the coordinates $\{x(\rho) \mid \rho > 0\}$ then the image of this point under σ has the coordinates $\{x'(\rho) \mid \rho > 0\}$, where $x'(\rho)$ is determined by

$$\sigma(n_{\sigma_T^{-1}(W_0)}) c_\sigma b c_\sigma \sigma(n_{\sigma_T^{-1}(W_0)}^{-1})^{-1} = b' = n_0 t_0 \prod_{\rho \in \Sigma^+} \exp(x'(\rho) X_\rho), \tag{2.9}$$

for some $n_0 \in N_+ =< U_\rho \mid \rho > 0 >$, where Σ^+ denotes the set of roots of T_+ which are positive with respect to B_-. Using Chevalley's commutator formula, this formula for $x'(\rho)$ will be used for explicit calculations below.

Lemma 2.10 [H2, pf of Lemma 5.3]**.** *(a) Let α denote an arbitrary simple positive root of G, an arbitrary quasi-split group. If $(u, (B(W))_{W \in \mathscr{W}}) \in E_{\Delta - \alpha}$ then every such Borel subgroup $B(W)$ lies in the same line ℓ_β in the Dynkin curve for G. Here $E_{\Delta-\alpha}$ is a 1-regular Richardson branch of $\phi^{-1}(1)$ on Y_C as in §3 below.*

(b) The action of Galois on the points of $E_{\Delta-\alpha}$ is determined by the action on the lines in the Dynkin curve.

Now we restrict to the case $G = G_2$.

Recall the Dynkin curve for G_2 [St]:

Possible Galois actions on this diagram include:
(i) exchanging ℓ'_β with ℓ'''_β, leaving ℓ''_β fixed, and flipping ℓ_α;
(ii) flipping ℓ'_β, ℓ'''_β, ℓ''_β and fixing ℓ_α;
(iii) cyclically permuting the lines ℓ'_β, ℓ'''_β, ℓ''_β and "rotating" ℓ_α.

Label and order the positive roots of the maximal torus $T_+ \subset B_-$ of G_2 as

$$\alpha \text{ (short)}, \beta \text{ (long)}, \gamma = \alpha + \beta, \delta = 2\alpha + \beta, \varepsilon = 3\alpha + \beta, \zeta = 3\alpha + 2\beta.$$

We have

$$\theta_\delta(r)\theta_\gamma(s) = \theta_\gamma(s)\theta_\delta(r)\theta_\zeta(3rs),$$

so

$$\begin{aligned}\theta_\varepsilon(x)^{\theta_{-\alpha}(z)} &= \theta_\varepsilon(x)\theta_\delta(c_1xz)\theta_\gamma(c_2xz^2)\theta_\beta(c_3xz^3)\\ &= \theta_\varepsilon(x)\theta_\beta(c_3xz^3)\theta_\delta(c_1xz)\theta_\gamma(c_2xz^2)\\ &= \theta_\beta(c_3xz^3)\theta_\gamma(c_2xz^2)\theta_\delta(c_1xz)\theta_\varepsilon(x)\theta_\zeta(c_4x^2z^3),\end{aligned}$$

where the c_i's are computable constants. Likewise,

$$\begin{aligned}\theta_\delta(x)^{\theta_{-\alpha}(z)} &= \theta_\delta(x)\theta_\gamma(d_1xz)\theta_\beta(d_2xz^2)\\ &= \theta_\beta(d_2xz^2)\theta_\gamma(d_1xz)\theta_\delta(x)\theta_\zeta(d_3x^2z),\\ \theta_\gamma(x)^{\theta_{-\alpha}(z)} &= \theta_\gamma(x)\theta_\beta(exz) = \theta_\beta(exz)\theta_\gamma(x),\\ \theta_\beta(x)^{\theta_{-\alpha}(z)} &= \theta_\beta(x),\end{aligned}$$

and $\theta_\alpha(x)^{\theta_{-\alpha}(z)} = \theta_{-\alpha}(y_\alpha)t\theta_\alpha(x)$, for some y_α depending on x, z, and some t belonging to a maximal torus.

In general, if $\sigma \in \Gamma$ acts on $T_G(\overline{F})$ pointwise as as simple reflection s_ρ, some $\rho \in \Delta$, then we write $\sigma = \sigma_\rho$. This amounts to taking $\sigma(n_{\sigma_T^{-1}(W_0)}) = \theta_\rho(z)$, where $z = \sigma(z(\rho)z_1(W_0,\rho))$ [H2, §3]. We will compute the Galois action on the coordinate of a point P in $Y''(B_+,B_-)$, where we assume that $B_+ \cap B_- = T_+$ is defined over F and $c_\sigma = 1$ in (2.9). (We will allow c_σ to be a reflection below, see (2.16-17).)

Taking $\sigma = \sigma_\alpha$, we have

$$\prod_{\rho>0}\theta_\rho(x(\rho))^{\theta_{-\alpha}(z)} = \theta_{-\alpha}(y_\alpha)t\prod_{\rho>0}\theta_\rho(\phi_\rho^\alpha(\underline{x},z)), \tag{2.11}$$

where $\underline{x} = (x(\alpha),\cdots,x(\zeta))$,

$$\begin{aligned}\phi_\alpha^\alpha(\underline{x},z) &= x(\alpha),\\ \phi_\beta^\alpha(\underline{x},z) &= x(\beta) + ezx(\gamma) + d_2z^2x(\delta) + c_3z^3x(\varepsilon),\\ \phi_\gamma^\alpha(\underline{x},z) &= x(\gamma) + d_1zx(\delta) + c_2z^2x(\varepsilon),\\ \phi_\delta^\alpha(\underline{x},z) &= x(\delta) + c_2zx(\varepsilon),\\ \phi_\varepsilon^\alpha(\underline{x},z) &= x(\varepsilon),\\ \phi_\zeta^\alpha(\underline{x},z) &= x(\zeta) + d_3zx(\delta)^2 + c_4z^3x(\varepsilon)^2 + 3c_2z^2x(\delta)x(\varepsilon).\end{aligned}$$

Therefore, if $c_{\sigma_\alpha} = 1$, we have

$$\sigma_\alpha : x(\rho) \longmapsto \phi_\rho^\alpha(\underline{x},z), \tag{2.12}$$

where $z = \sigma_\alpha(z(\alpha)z_1(W_0,\alpha))$.

Take $\sigma = \sigma_\beta$ and note

$$\begin{aligned}\theta_\gamma(r)\theta_\alpha(s) &= \theta_\alpha(s)\theta_\gamma(r)\theta_\delta(2rs)\theta_\varepsilon(3rs^2)\theta_\zeta(3r^2s),\\ \theta_\beta(r)\theta_\alpha(s) &= \theta_\alpha(s)\theta_\beta(r)\theta_\gamma(rs)\theta_\delta(r^2s)\theta_\varepsilon(r^3s)\theta_\zeta(r^3s^2).\end{aligned}$$

We have, by Chevalley's commutator formula,

$$\theta_\alpha(x)^{\theta_{-\beta}(z)} = \theta_\alpha(x),$$

$$\theta_\beta(x)^{\theta_{-\beta}(z)} = \theta_{-\beta}(y_\beta)t\theta_\beta(x), \quad (t \in T_G,\ \alpha(t) = 1+xz,\ y_\beta := \frac{-x^2z}{1+xz}),$$

$$\theta_\gamma(x)^{\theta_{-\beta}(z)} = \theta_\gamma(x)\theta_\alpha(czx)$$
$$= \theta_\alpha(czx)\theta_\gamma(x)\theta_\delta(2czx^2)\theta_\varepsilon(3c^2z^2x^3)\theta_\zeta(3czx^3),$$
$$\theta_\delta(x)^{\theta_{-\beta}(z)} = \theta_\delta(x),$$
$$\theta_\varepsilon(x)^{\theta_{-\beta}(z)} = \theta_\varepsilon(x),$$
$$\theta_\zeta(x)^{\theta_{-\beta}(z)} = \theta_\zeta(x)\theta_\varepsilon(dzx) = \theta_\varepsilon(dzx)\theta_\zeta(x).$$

Therefore,

$$\prod_{\rho>0}\theta_\rho(x(\rho))^{\theta_{-\beta}(z)} = t\prod_{\rho>0}\theta_\rho(\phi_\rho^\beta(\underline{x},z)), \tag{2.13}$$

where

$$\phi_\alpha^\beta(\underline{x},z) = x(\alpha)(1+x(\beta)z)+cx(\gamma)z,$$
$$\phi_\beta^\beta(\underline{x},z) = x(\beta),$$
$$\phi_\gamma^\beta(\underline{x},z) = x(\gamma)+czx(\beta)x(\gamma),$$
$$\phi_\delta^\beta(\underline{x},z) = x(\delta)+2czx(\gamma)^2+c^2x(\beta)x(\gamma)^2z^2,$$
$$\phi_\varepsilon^\beta(\underline{x},z) = x(\varepsilon)+dzx(\zeta)+3c^2x(\gamma)^3z^2+c^3x(\beta)x(\gamma)^3z^3,$$
$$\phi_\zeta^\beta(\underline{x},z) = x(\zeta)+3zx(\gamma)x(\delta)+9czx(\gamma)^3+c^3z^3x(\beta)^2x(\gamma)^3.$$

Therefore, if $c_{\sigma_\beta}=1$, we have

$$\sigma_\beta : x(\rho) \longmapsto \phi_\rho^\beta(\underline{x},z), \tag{2.14}$$

where $z = \sigma_\beta(z(\beta)z_1(W_0,\beta))$.

Formulas (2.12) and (2.14) help to determine the set of F-rational points of Y_C. We also need to compute the effect of conjugating b by c_σ. For this we recall a basic lemma [Car]:

Lemma 2.15. *Let $\phi_\rho : SL(2)\to G$ be the homomorphism associated to the root ρ and let $\theta_\rho : \mathbb{G}_a \to U_\rho$ denote the root group embedding, so $\theta_\rho(x) = \phi_\rho(\begin{pmatrix}1 & x\\ 0 & 1\end{pmatrix})$. Let $s_\rho := s_\rho(1)$ where*

$$s_\rho(x) = \theta_\rho(x)\theta_{-\rho}(-x^{-1})\theta_\rho(x) = \phi_\rho(\begin{pmatrix}0 & x\\ -1/x & 0\end{pmatrix}).$$

Then

$$s_\rho\theta_\sigma(x)s_\rho^{-1} = \theta_{w_\rho(\sigma)}(\eta_{\rho,\sigma}x),$$

where $\eta_{\rho,\sigma}$ is as in [Car, pp 94-96] *and w_ρ is the reflection of the Weyl group associated to ρ.*

As a consequence of this, we may write

$$s_\alpha\cdot\prod_{\rho>0}\theta_\rho(x(\rho))\cdot s_\alpha^{-1}$$
$$= \theta_{-\alpha}(x'(\alpha))\prod_{\rho>0,\rho\neq\alpha}\theta_\rho(x'(\rho)), \tag{2.16}$$

where $x'(\alpha) = \eta_{\alpha,\alpha}x(\alpha)$, $x'(\beta) = \eta_{\alpha,\varepsilon}x(\varepsilon)$, $x'(\gamma) = \eta_{\alpha,\delta}x(\delta)$, $x'(\delta) = \eta_{\alpha,\gamma}x(\gamma)$, $x'(\varepsilon) = \eta_{\alpha,\beta}x(\beta)$, and

$$x'(\zeta) = \eta_{\alpha,\zeta}x(\zeta)+f_1x(\delta)x(\gamma)+f_2x(\beta)x(\varepsilon),$$

where f_1, f_2 are constants. Likewise, we may write

$$\begin{aligned} & s_\beta \cdot \prod_{\rho>0} \theta_\rho(x(\rho)) \cdot s_\beta^{-1} \\ & = \theta_{-\beta}(x'(\beta)) \prod_{\rho>0, \rho\neq\beta} \theta_\rho(x'(\rho)), \end{aligned} \tag{2.17}$$

where

$$x'(\alpha) = \eta_{\alpha,\gamma} x(\gamma) + f_3 x(\alpha) x(\beta)$$

$x'(\beta) = \eta_{\alpha,\beta} x(\beta)$, $x'(\gamma) = \eta_{\alpha,\alpha} x(\alpha)$,

$$x'(\delta) = \eta_{\alpha,\delta} x(\delta) + f_4 x'(\alpha) x'(\gamma),$$

$x'(\varepsilon) = \eta_{\alpha,\zeta} x(\zeta)$, and

$$x'(\zeta) = \eta_{\alpha,\varepsilon} x(\varepsilon).$$

§ 3 Construction of regular patches

This section is a preliminary step to constructing the regular variety Y_C for G_2.

Let G denote an arbitrary connected semisimple group over $\mathscr{O}_F$. Let $\chi : G \to T_G/\Omega$ denote the adjoint quotient [Slo]. There is an open patch

$$Y''(B_+, B_-) \times N_+ \subset \chi^{-1}(C) \times S''(B_+, B_-) \times N_+ \subset G \times S''(B_+, B_-) \times N_+$$

of Y'' which is isomorphic to an open set of the product $\mathbb{A}^p \times Z$, where $Z \subset \mathbb{A}^{2p+1}$ is a "monomial" $p+1$ dimensional affine variety having coordinates

$$w(\rho), \quad (\rho \in \Sigma_G^+ - \Delta), \quad z(\alpha) \quad (\alpha \in \Delta), \quad x(\rho) \quad (\rho \in \Sigma_G^+), \quad \lambda,$$

(we set $w(\alpha) = 1,\ \alpha \in \Delta$), defined by the relations: for $\lambda \neq 0$, $\rho \in \Sigma_G^+$, we have

$$\lambda w(\rho) = x(\rho) \prod_{\alpha\in\Delta} z(\alpha)^{m(\alpha)}, \tag{3.1}$$

where $m(\alpha) = m_\rho(\alpha) \geq 0$ is the integral coefficient in $\rho = \sum_{\alpha\in\Delta} m(\alpha)\alpha$. Here $\Delta \subset \Sigma_G$ denotes a basis of simple roots of T_+ in B_-. There is a surjection $Y''(B_+) \to Z$ which allows us to pull-back open subvarieties $Z_U \subset Z$ to open subvarieties $Y''_U \subset Y''$. Moreover, if $\rho = \alpha$ is simple then $\lambda = x(\alpha)z(\alpha)$. This variety Z is regular except possibly for a singularity contained in the hypersurface $\lambda = 0$.

We first claim that, for the case $G = G_2$, the singularities of $Y''(B_+, B_-)$, if any, lie in

$$K: \quad x(\alpha) = x(\beta) = x(\gamma) = z(\alpha) = w(\zeta) = 0$$

or in

$$K_0: \quad x(\alpha) = x(\beta) = x(\gamma) = x(\delta) = x(\varepsilon) = x(\zeta) = w(\zeta) = 0.$$

The points in $Y''(B_+, B_-)$ with coordinates in the latter set all map to the identity in G under π, hence correspond to the trivial unipotent orbit. Therefore, blowing up Z along K_0 creates exceptional divisors associated with the trivial orbit, which do not concern us here.

To prove this claim, we exhibit 5 regular patches, give generators of the local coordinate ring, and express the remaining coordinates as regular functions of the generators on that patch.

$x(\alpha) \neq 0$: Generators: $x(\alpha)$, $x(\beta)$, $z(\beta)$, $x(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\beta)z(\beta)$, $z(\alpha) = x(\beta)z(\beta)/x(\alpha)$, $w(\gamma) = x(\gamma)z(\beta)/x(\alpha)$, $w(\delta) = x(\delta)z(\alpha)z(\beta)/x(\alpha)$, $w(\varepsilon) = x(\varepsilon)z(\alpha)^2 z(\beta)/x(\alpha)$, $w(\zeta) = x(\zeta)z(\alpha)^2 z(\beta)^2/x(\alpha)$.

$x(\beta) \neq 0$: Generators: $x(\alpha)$, $x(\beta)$, $z(\alpha)$, $x(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\alpha)z(\alpha)$, $z(\beta) = x(\alpha)z(\alpha)/x(\beta)$, $w(\gamma) = x(\gamma)z(\alpha)/x(\beta)$, $w(\delta) = x(\delta)z(\alpha)^2/x(\beta)$, $w(\varepsilon) = x(\varepsilon)z(\alpha)^3/x(\beta)$, $w(\zeta) = x(\zeta)z(\alpha)^5x(\alpha)^2/x(\beta)^3$.

$x(\gamma) \neq 0$: Generators: $x(\alpha)$, $x(\beta)$, $x(\gamma)$, $w(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\alpha)x(\beta)w(\gamma)/x(\gamma)$, $z(\alpha) = x(\beta)w(\gamma)/x(\gamma)$, $z(\beta) = x(\alpha)w(\gamma)/x(\gamma)$, $w(\delta) = x(\delta)w(\gamma)^2x(\beta)/x(\gamma)^2$, $w(\varepsilon) = x(\varepsilon)w(\gamma)^3x(\beta)^2/x(\gamma)^3$, $w(\zeta) = x(\zeta)x(\alpha)x(\beta)^2w(\gamma)^4/x(\gamma)$.

$w(\zeta) \neq 0$: Generators: $z(\alpha)$, $z(\beta)$, $w(\gamma)$, $w(\delta)$, $w(\varepsilon)$, $w(\zeta)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\zeta)z(\alpha)^3z(\beta)^2/w(\zeta)$, $x(\alpha) = x(\zeta)z(\alpha)^2z(\beta)^2/w(\zeta)$, $x(\beta) = x(\zeta)z(\alpha)^3z(\beta)/w(\zeta)$, $x(\gamma) = w(\gamma)x(\zeta)z(\alpha)^2z(\beta)/w(\zeta)$, $x(\delta) = w(\delta)x(\zeta)z(\alpha)z(\beta)/w(\zeta)$, $x(\varepsilon) = w(\varepsilon)x(\zeta)z(\beta)/w(\zeta)$.

$z(\alpha) \neq 0$ (assuming $x(\varepsilon) \neq 0$): Generators: $z(\alpha)$, $z(\beta)$, $x(\beta)$, $w(\gamma)$, $w(\delta)$, $w(\varepsilon)$, $x(\zeta)$. Remaining coordinates: $\lambda = x(\beta)z(\beta)$, $x(\alpha) = x(\beta)z(\beta)/z(\alpha)$, $x(\gamma) = w(\gamma)x(\beta)/z(\alpha)$, $x(\delta) = w(\delta)x(\beta)/z(\alpha)^2$, $x(\varepsilon) = w(\varepsilon)x(\beta)/z(\alpha)^3$, $w(\zeta) = \frac{x(\zeta)}{x(\varepsilon)}w(\varepsilon)z(\beta)/z(\alpha)^3$. We remark that the assumption $x(\varepsilon) \neq 0$ may be replaced by $x(\delta) \neq 0$ or $x(\zeta) \neq 0$ without affecting regularity (though of course the generators, etc., will change).

The above claim follows.

As remarked above, the "interesting" singularities of Z lie in the subvariety K above, not in K_0 since the exceptional divisor associated to the blow-up along K_0 corresponds to the trivial orbit, whose matching has be deduced from the work of Rogawski, Howe, and Harish-Chandra [Ro].

By Hironaka's embedded resolution of singularities theorem (see also Mumford's semi-stable reduction theorem [K]), we may obtain a smooth variety $\tilde{Z} \to Z$, proper over Z, by blowing up a finite number of times, beginning with K as the center of the first blow up. Moreover, the exceptional divisor E_Z describing $\lambda = 0$ in $\tilde{Z}$ will have normal crossings. Let $\tilde{Y''} \to Y''$ be the corresponding resolution of Y'' with exceptional divisor $E_{Y''} \subset \tilde{Y''}$; the branches of $E_{Y''}$ are reduced and cross normally. This can (and has) been done explicitly. However, as the details need not concern us here they will be omitted.

Let G be an arbitrary connected semisimple group over an algebraically closed field again. Let $\Delta = \{\alpha_1, ..., \alpha_\ell\}$, which we sometimes identify with $\{1, ..., \ell\}$, $\ell := rk\, G$. Let $S_1, S_2 \subset \{1, ..., \ell\}$ be disjoint sets, and if they are non-empty then let $i_{min} := min_{i \in S_1} i$, $j_{min} := min_{j \in S_2} j$, and similarly for i_{max}, j_{max}. If S_1, S_2 are non-empty, not only do we assume they are disjoint but also that (i) $i_{max} < j_{min}$ or $j_{max} < i_{min}$ and (ii) if $S_2 \neq \emptyset$ then $S_1 \cup S_2 \neq \Delta$.

For each map $\sigma : S_2 \to \Sigma_G^+ - \Delta$, define the subset $Z_{S_1,S_2,\sigma} \subset Z$ by
(3.2a) $z(\alpha) \neq 0$, for all $\alpha \notin S_1 \cup S_2$,
(3.2b) $x(\alpha) \neq 0$, for all $\alpha \in S_1$,
(3.2c) $x(\sigma(j)) \neq 0$, for all $j \in S_2$,
(3.2d) $w(\sigma(j)) \neq 0$, for all $j \in S_2$.
These are **fundamental patches** of Z in the sense of [H1]. This collection of patches depends, a priori, on the identification of Δ with $\{1, ..., \ell\}$ above. For $G = G_2$, they will turn out to have the nice property that the divisor $\lambda = 0$ on Z will only meet $Z_{S_1,S_2,\sigma}$ along a single irreducible branch which we will denote by $E_{S_1,S_2,\sigma}$, possibly occurring with some multiplicity $a(E_{S_1,S_2,\sigma})$. The $E_{S_1,S_2,\sigma}$ are called the **fundamental branches** of $\lambda = 0$ on Z. If $S_2 = \{\rho\}$ is a singleton and $\tau = \sigma(\rho)$ then for brevity we write $E_{S_1,\rho,\tau}$ in place of $E_{S_1,S_2,\sigma}$. If $S_2 = \emptyset$ the we abbreviate this patch by Z_{S_1} and call it a **Richardson patch**. Similarly for E_{S_1}, which we call a **Richardson branch**. (I call the Z_{S_1} "Richardson patches", since by Proposition 9.1 of Ch II of [H1] they will lead to divisors E of Y for which $\pi(E)$ is the Zariski closure of a Richardson orbit.) Notice that the **regular branch** $E_\emptyset$ ($S_1 = S_2 = \emptyset$) satisfies $\pi(E_\emptyset) = \overline{\mathcal{O}}_{reg} = V$ where V denotes the unipotent variety of G and $\mathcal{O}_{reg}$

denotes the regular orbit of G. Since $E_\emptyset \subset Y_C$ is regular, the map $\pi : E_\emptyset \to V$ yields a resolution of singularities similar to the Springer resolution [Slo], [Sp], [St].

Fix a map $\sigma : S_2 \to \Sigma_G^+ - \Delta$ satisfying
(3.3a) for $j, j' \in S_2$, the simple root expansion of $\sigma(j)$ and $\sigma(j')$ contain no common simple roots unless $j = j'$ or j, j' are consecutive.
(3.3b) If j, j' are consecutive in S_2 then

$$\sigma(j) = \sum_{k=j}^{j'} m_{jk}\alpha_k,$$

for some integers $m_{jk} \geq 0$.
(3.3c) There is an $a \geq 1$ for which $\sigma(j_{max}) = a\alpha_{j_{max}} + \sum_{\alpha \notin S_1 \cup S_2} m(\alpha)\alpha$, for some integers $m(\alpha) \geq 0$.

Such a map will be called **essential** (in the sense that there are lots of other possible patches $Z_{S_1,S_2,\sigma}$ but the patches which are not essential are expected to be redundant in the covering of Z by patches up to a set of codimension 2). The corresponding fundamental patches are called **essential**. This terminology is admittedly a bit optimistic: there are, in general, many extraneous essential patches. Therefore this definition is only provisional. In any case, these patches generalize the sequences $\gamma(i, j), \ldots$ of roots used in the paper [H2].

Notation. *The notation for these divisors is not exactly the same as that of* [H1], [H2], *... , though the notation for the patches seems to agree. Hales' E_S is usually my $E_{\Delta - S}$.*

Rather than discuss these patches in greater generality than is necessary for our purposes here, we merely state the following

Conjecture 3.4. *If G is not of type D_n, E_n, F_4 then Z is covered by the $Z_{S_1,S_2,\sigma}$, except for a subset of codimension ≥ 2. Moreover, the $Z_{S_1,S_2,\sigma}$ are all regular and Z is regular in codimension one.*

In the case of G_2 this follows from the next two lemmas. The conjecture is known to be true in the case of A_n and C_n [H1], [H2], and false in the case of D_n.

Lemma 3.5 (Hales [H1]**).** *For $S_1 \subset \Delta$ non-empty, Z_{S_1} is regular, with generators λ, $z(\alpha)$ $(\alpha \in \Delta - S_1)$, $x(\alpha)$ $(\alpha \in S_1)$, $x(\rho)$ $(\rho \in \Sigma_G^+ - \Sigma_{S_1}^+)$, $w(\rho)$ $(\rho \in \Sigma_{S_1}^+)$, where $\Sigma_{S_1}^+$ denotes the subroot system generated by $S_1 \subset \Delta$. It is therefore $p+1$ dimensional and the divisor $\lambda = 0$ occurs with multiplicity one.*

It is also easy to prove the conjecture when S_1 is empty and S_2 is a singleton. For brevity, let m_α denote the multiplicity with which $z(\alpha) = 0$ occurs in $\lambda = 0$ and let $\Delta_{\lambda=0} := \{\beta \in \Delta \mid m_\beta \neq 0\}$.

Lemma 3.6. *For any connected semi-simple group G, any simple root $\alpha \in \Delta$, and any non-simple positive root $\sigma \in \Sigma_G^+ - \Delta$, if $Z_{\emptyset,\alpha,\sigma}$ is an essential patch then $Z_{\emptyset,\alpha,\sigma}$ is regular and its coordinate ring has generators*

$$\begin{aligned} &z(\beta) \ \ (\beta \in \Delta), \\ &x(\rho) \quad \text{if } \rho > \sigma, \\ &w(\rho) \quad \text{if } \rho \leq \sigma \text{ and } \rho \notin \Delta. \end{aligned}$$

Moreover, the divisor $\lambda = 0$ occurs as $z(\alpha) = 0$ with multiplicity m, where $m \geq 1$ is the integral coefficient of α in the simple root expansion of σ.

Proof. Let $\sigma = m\alpha + \sum_{\beta \neq \alpha} m_\beta \beta$ and notice that we may assume that $\sigma \notin \Delta$ for otherwise the patch is Richardson.

Recall the definition of $Z_{\emptyset,S_2,\sigma}$: $z(\beta)\neq 0$ for $\beta\neq\alpha$ simple, $x(\sigma)\neq 0$, and $w(\sigma)\neq 0$. By (3.3c) and our hypothesis, $\lambda w(\sigma)=x(\sigma)z(\alpha)^m\prod_{\beta\neq\alpha}z(\beta)^{m_\beta}$, it is clear that $\lambda=0$ occurs as $z(\alpha)=0$ with multiplicity m_α. Along $\lambda=0$ the other $z(\beta)'s$ are non-zero, and the $\{z(\beta)\mid\beta\in\Delta\}$ give us ℓ independent coordinates. We need $p+1-\ell$ more having the property that the remaining coordinates are regular functions of these along $\lambda=0$. Suppose $\rho\leq\sigma$, so $\sigma=\rho+\sum_\beta m(\beta)\beta$. We have

$$x(\rho)=\frac{x(\sigma)w(\rho)}{w(\sigma)}\prod_\beta z(\beta)^{m(\beta)},$$

and therefore we want to choose the $w(\rho)$ with $\rho\leq\sigma$, $\rho\notin\Delta$, as generators. Likewise, we want to choose the $x(\rho)$ with $\rho\geq\sigma$ as generators since a similar line of reasoning applied to the quotients

$$\frac{w(\rho)}{w(\sigma)}=\frac{x(\rho)}{x(\sigma)}\prod_\beta z(\beta)^{m_\rho(\beta)-m_\sigma(\beta)}.$$

This yields the claim. □

Lemma 3.7. *For any connected semisimple group, the variety Z is normal and Cohen-Macauley. Moreover, Z is a set-theoretic complete intersection.*

Proof. Z is "monomial" and hence has a toroidal embedding [K, Ch I]. From the definition of saturated [K, Ch I, Definition 2] and the equations (3.1) defining Z, we see that any toroidal embedding $T\hookrightarrow Z$ corresponds to a saturated sub-semigroup of $X^*(T)=Hom_{alg.-grps.}(T,\mathbb{G}_m)$. Z is therefore normal by [K, Ch I, Theorem 1] and Cohen-Macauley by [K, Ch I, Theorem 14]. The fact that Z is a set-theoretic complete intersection follows from the definition and counting the number of relations defined by (3.1). □

§ 4 Regularity of patches for $G=G_2$

It is not difficult to write all these patches of §3 down and show that they are regular for G_2. Recall $\Sigma_G^+=\{\alpha,\beta,\gamma:=\alpha+\beta,\delta:=2\alpha+\beta,\varepsilon:=3\alpha+\beta,\zeta:=3\alpha+2\beta\}$, with α short and β long, so $p=6$ and $\dim Z=7$. We have variables

$$\begin{aligned}&x(\alpha),x(\beta),x(\gamma),x(\delta),x(\varepsilon),x(\zeta)\\ &z(\alpha),z(\beta),w(\gamma),w(\delta),w(\varepsilon),w(\zeta)\\ &\lambda,\end{aligned}\tag{4.1}$$

subject to the usual relations (3.1).

With B_+ choosen as in §3, we define $\omega_{Y''(B_+)}$ to be the G-invariant form

$$\omega_{Y''(B_+)}=d\lambda\wedge\wedge dx(\alpha)\wedge dx(\beta)\wedge dx(\gamma)\wedge dx(\delta)\wedge dx(\varepsilon)\wedge dx(\zeta)\wedge dv,$$

where dv is a form on N_+. We use this to compute the Igusa constants β in this section as in [H2].

As we know (from lemma 3.5), the Richardson patches Z_{S_1} are regular. Now let us consider the remaining patches. Since $S_1\cup S_2\neq\Delta$ if $S_2\neq\emptyset$, it is easy to list them all: $S_1=\emptyset$, $S_2=\{\alpha\}$, where $\sigma(\alpha)=\alpha,\gamma,\delta,\varepsilon,\zeta$, or $S_1=\emptyset$, $S_2=\{\beta\}$, where $\sigma(\beta)=\beta,\gamma,\delta,\varepsilon,\zeta$. This may be deduced from the discussion above. We next list the generators of the local coordinate ring in each case:
$S_2=\{\alpha\},\sigma(\alpha)=\alpha$: Same as $Z_{\{\alpha\}}$. The divisor $\lambda=0$ occurs with multiplicity one.
$S_2=\{\alpha\},\sigma(\alpha)=\gamma$: Generators: $z(\alpha),z(\beta),x(\gamma),w(\gamma),x(\delta),x(\varepsilon),x(\zeta)$. Regular at $\lambda=0$. The divisor $\lambda=0$ occurs with multiplicity one (represented by $z(\alpha)=0$).
$S_2=\{\alpha\},\sigma(\alpha)=\delta$: Generators: $z(\alpha),z(\beta),w(\gamma),x(\delta),w(\delta),x(\varepsilon),x(\zeta)$. Regular at $\lambda=0$. The divisor $\lambda=0$ occurs (as $z(\alpha)=0$) with multiplicity two.

$S_2 = \{\alpha\}, \sigma(\alpha) = \varepsilon$: Generators: $z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), x(\varepsilon), x(\zeta)$. Regular at $\lambda = 0$. The divisor $\lambda = 0$ occurs (as $z(\alpha) = 0$) with multiplicity three.
$S_2 = \{\alpha\}, \sigma(\alpha) = \zeta$: Generators: $z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), w(\zeta), x(\zeta)$. Regular at $\lambda = 0$. The divisor $\lambda = 0$ occurs (as $z(\alpha) = 0$) with multiplicity three.
$S_2 = \{\beta\}, \sigma(\beta) = \beta$: Same as $Z_{\{\beta\}}$. The divisor $\lambda = 0$ occurs with multiplicity one.
$S_2 = \{\beta\}, \sigma(\beta) = \gamma$: Generators: $z(\alpha), z(\beta), x(\gamma), w(\gamma), x(\delta), x(\varepsilon), x(\zeta)$. Regular at $\lambda = 0$. The divisor $\lambda = 0$ occurs with multiplicity one (represented by $z(\beta) = 0$).
$S_2 = \{\beta\}, \sigma(\beta) = \delta$: Generators: $z(\alpha), z(\beta), w(\gamma), x(\delta), w(\delta), x(\varepsilon), x(\zeta)$. Regular at $\lambda = 0$. The divisor $\lambda = 0$ occurs (as $z(\beta) = 0$) with multiplicity one.
$S_2 = \{\beta\}, \sigma(\beta) = \varepsilon$: Generators: $z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), x(\varepsilon), x(\zeta)$. Regular at $\lambda = 0$. The divisor $\lambda = 0$ occurs (as $z(\beta) = 0$) with multiplicity one.
$S_2 = \{\beta\}, \sigma(\beta) = \zeta$: Generators: $z(\alpha), z(\beta), w(\gamma), w(\delta), w(\varepsilon), w(\zeta), x(\zeta)$. Regular at $\lambda = 0$. The divisor $\lambda = 0$ occurs (as $z(\beta) = 0$) with multiplicity two.

There is some redundancy here since $E_{\emptyset,\{\beta\},\sigma} = E_{\{\beta\}}$ if $\sigma(\beta) = \beta, \gamma, \delta, \varepsilon$.

We denote the (irreducible) branch of $\{\lambda = 0\} \subset Z$ occuring in $Z_{S_1,S_2,\sigma}$ by $E_{S_1,S_2,\sigma}$. The same notation will be used for the corresponding branch in Y_U and for its Zariski closure in Y_1. To determine which of the κ-stable germs, if any, the divisor $E_{S_1,S_2,\sigma}$ contributes to by formula 0.6, we must compute the Igusa constant

$$\beta(\tilde{E}_{S_1,S_2,\sigma}) = \frac{b(\tilde{E}_{S_1,S_2,\sigma})}{a(\tilde{E}_{S_1,S_2,\sigma})} \tag{4.2}$$

where $\tilde{E}$ denotes the preimage of E in the desingularized variety Y_C.

We record two observations regarding the Igusa constants $\beta(E)$.
(I) First, if $\phi : V \to W$ is an isomorphism of smooth varieties which sends a divisor $E \subset V$ to a divisor $F = \phi(E) \subset W$ then $\beta(E) = \beta(F)$.
(II) Secondly, let V' denote a blow up of a variety V with exceptional divisor $E_{V'}$. Recall that for each irreducible branch E on V, the pull-back ξ'^* of the blow down map $\xi' : V' \to V$ satisfies $\xi'^*(E) = \tilde{E} + \delta E_{V'}$, where $\tilde{E}$ is the strict transform of E, and

$$\delta := \begin{cases} 1, & \text{if } E \text{ contains the center of the blow up,} \\ 0, & \text{otherwise.} \end{cases}$$

A repeated use of this global identity can be used to obtain local information about the Igusa constants on Z.

For example, if E denotes a branch of $\{\lambda = 0\} \subset Z$ and $\tilde{E}$ its strict transform in a desingularization $\tilde{Z}$, we have

$$\beta(\tilde{E}) = \beta(E),$$

provided E does not contain any of the images of the centers of the blow ups used to create $\tilde{Z}$. The only other divisors in $\{\lambda = 0\} \subset \tilde{Z}$ other than the strict transforms $\tilde{E}$ above are the exceptional divisors. To compute all the β-values of the branches of $\phi^{-1}(1)$ in Y_C we need only compute the β-values of the divisors $E_{S_1,S_2,\sigma}$ on Z and the branches of the exceptional divisors of the blow-up.

We digress to compute $\beta(E_{S_1,S_2,\sigma})$ on $Z_{S_1,S_2,\sigma}$. We need not compute the β-values for the exceptional divisors because of lemma 0.8 and the fact that all the needed F-rational $\mathcal{O}$-divisors can be accounted for among the $E_{S_1,S_2,\sigma}$.

If $S_2 = \emptyset$ then $\beta(E_{S_1,S_2,\sigma})$ has been computed in [H1, ch II, Proposition 9.1] so some of our calculations are covered by Hales' general calculations. For completeness, we list everything anyway.

Igusa forms on the fundamental patches of Y''. We pull-back the patch $Z_{S_1,S_2,\sigma}$ to $Y''(B_+,B_-)$ via the open immersion $Y''(B_+,B_-) \to Z$. Call this inverse image $Y''_{S_1,S_2,\sigma}$. The Igusa form ω_U on $U = Y''_{S_1,S_2,\sigma}$ is related to the Igusa form $\omega_{Z,U}$ on $Z_{S_1,S_2,\sigma}$ by $\omega_U = \omega_{Z,U} \wedge dv$, where dv is regular. We compute $\omega_{Z,U}$ below.

$Z_{\{\alpha,\beta\}}$: Here the multiplicity $a(E_{\{\alpha,\beta\}}) = 1$ and $b(E_{\{\alpha,\beta\}}) = 1$ since

$$\omega_{Z,U} = d\lambda \wedge dx(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta).$$

Therefore, $\beta(E_{\{\alpha,\beta\}}) = 1$.

Z_α: Here the multiplicity $a(E_\alpha) = 1$ and $b(E_\alpha) = 2$ since

$$\begin{aligned}\omega_{Z,U} &= d\lambda \wedge dx(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta) \\ &= -\frac{\lambda}{z(\beta)^2} d\lambda \wedge dx(\alpha) \wedge dz(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta).\end{aligned}$$

Therefore, $\beta(E_\alpha) = 2$.

Z_β: Here the multiplicity $a(E_\beta) = 1$ and $b(E_\beta) = 2$ since

$$\begin{aligned}\omega_{Z,U} &= d\lambda \wedge dx(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta) \\ &= -\frac{\lambda}{z(\alpha)^2} d\lambda \wedge dz(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta).\end{aligned}$$

Therefore, $\beta(E_\beta) = 2$.

$Z_{\emptyset,\alpha,\sigma},\ \sigma(\alpha) = \gamma$: Here the multiplicity $a(E_{\emptyset,\alpha,\sigma}) = 1$, by the above list. Also, $b(E_{\emptyset,\alpha,\sigma}) = 2$ since

$$\begin{aligned}\omega_{Z,U} &= d\lambda \wedge dx(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta) \\ &= c\frac{x(\gamma)^3 z(\beta) z(\alpha)}{w(\gamma)^4} dz(\alpha) \wedge dz(\beta) \wedge dw(\gamma) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta).\end{aligned}$$

Therefore, $\beta(E_{\emptyset,\alpha,\sigma}) = 2$.

$Z_{\emptyset,\alpha,\sigma},\ \sigma(\alpha) = \delta$: Here the multiplicity $a(E_{\emptyset,\alpha,\sigma}) = 2$, by the above list. Also, $b(E_{\emptyset,\alpha,\sigma}) = 6$ since

$$\begin{aligned}\omega_{Z,U} &= d\lambda \wedge dx(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta) \\ &= c\frac{x(\delta)^4 z(\beta) z(\alpha)^5}{w(\delta)^5} dz(\alpha) \wedge dz(\beta) \wedge dw(\gamma) \wedge dw(\delta) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta).\end{aligned}$$

Therefore, $\beta(E_{\emptyset,\alpha,\sigma}) = 3$.

$Z_{\emptyset,\alpha,\sigma},\ \sigma(\alpha) = \varepsilon$: Here the multiplicity $a(E_{\emptyset,\alpha,\sigma}) = 3$, by the above list. Also, $b(E_{\emptyset,\alpha,\sigma}) = 10$ since

$$\begin{aligned}\omega_{Z,U} &= d\lambda \wedge dx(\alpha) \wedge dx(\beta) \wedge dx(\gamma) \wedge dx(\delta) \wedge dx(\varepsilon) \wedge dx(\zeta) \\ &= c\frac{x(\varepsilon)^5 z(\beta) z(\alpha)^9}{w(\delta)^6} dz(\alpha) \wedge dz(\beta) \wedge dw(\gamma) \wedge dw(\delta) \wedge dw(\varepsilon) \wedge dx(\varepsilon) \wedge dx(\zeta).\end{aligned}$$

Therefore, $\beta(E_{\emptyset,\alpha,\sigma}) = 10/3$.

$Z_{\emptyset,\alpha,\sigma},\ \sigma(\alpha)=\zeta$: Here the multiplicity $a(E_{\emptyset,\alpha,\sigma})=3$, by the above list. Also, $b(E_{\emptyset,\alpha,\sigma})=11$ since

$$\begin{aligned}\omega_{Z,U} &= d\lambda\wedge dx(\alpha)\wedge dx(\beta)\wedge dx(\gamma)\wedge dx(\delta)\wedge dx(\varepsilon)\wedge dx(\zeta)\\ &= c\frac{x(\zeta)^6 z(\beta)^7 z(\alpha)^{10}}{w(\delta)^7} dz(\alpha)\wedge dz(\beta)\wedge dw(\gamma)\wedge dw(\delta)\wedge dw(\varepsilon)\wedge dw(\zeta)\wedge dx(\zeta).\end{aligned}$$

Therefore, $\beta(E_{\emptyset,\alpha,\sigma})=11/3$.

$Z_{\emptyset,\beta,\sigma},\ \sigma(\beta)=\zeta$: Here the multiplicity $a(E_{\emptyset,\beta,\sigma})=2$, by the above list. Also, $b(E_{\emptyset,\beta,\sigma})=8$ since

$$\begin{aligned}\omega_{Z,U} &= d\lambda\wedge dx(\alpha)\wedge dx(\beta)\wedge dx(\gamma)\wedge dx(\delta)\wedge dx(\varepsilon)\wedge dx(\zeta)\\ &= c\frac{x(\zeta)^6 z(\beta)^7 z(\alpha)^{10}}{w(\delta)^7} dz(\alpha)\wedge dz(\beta)\wedge dw(\gamma)\wedge dw(\delta)\wedge dw(\varepsilon)\wedge dw(\zeta)\wedge dx(\zeta).\end{aligned}$$

Therefore, $\beta(E_{\emptyset,\beta,\sigma})=4$.

We summarize this in the following table:

(4.3)

S_1	S_2	σ	$\beta(E_{S_1,S_2,\sigma})-1$
α,β	$\emptyset$	$------$	0
α	$\emptyset$	$------$	1
β	$\emptyset$	$------$	1
$\emptyset$	$\emptyset$	$------$	trivial
$\emptyset$	α	α	1
$\emptyset$	α	γ	1
$\emptyset$	α	δ	2
$\emptyset$	α	ε	$7/3$
$\emptyset$	α	ζ	$8/3$
$\emptyset$	β	β	1
$\emptyset$	β	γ	1
$\emptyset$	β	δ	1
$\emptyset$	β	ε	1
$\emptyset$	β	ζ	3.

As was remarked above, there is some redundancy here... .

Let $H_1=PGL(3)$ and $H_2=(SL(2)\times SL(2))/(\pm 1)$.

Lemma 4.4. *In the notation of Lemmas 3.5 and 3.6, we have*

$$\beta(E_S)=\frac{b(E_S)}{a(E_S)}=1+|\Sigma_S^+|=1+\frac{dim(Cent_G(u))-2}{2},$$

where u is an element of the Richardson orbit $\mathcal{O}_S$ associated to the parabolic P_S and the divisor E_S is the Richardson divisor contained in the patch Z_S. The Igusa constants for

$$E_{\mathcal{O}}=\begin{cases}E_{\emptyset,\alpha,\delta}, & \mathcal{O}\ \textit{is 2-regular},\\ E_{\emptyset,\beta,\zeta}, & \mathcal{O}\ \textit{is 3-regular},\end{cases}$$

are given by

$$\beta(E_{\mathcal{O}})=\frac{b(E_{\mathcal{O}})}{a(E_{\mathcal{O}})}=1+|\Sigma_{H_i}^+|=1+\frac{dim(Cent_G(u))-2}{2},$$

where $i = 1,2$ is determined by the orbit as indicated below and u is an element of the induced orbit

$$\mathcal{O} = \begin{cases} \mathrm{Ind}_{H_2}^G(1), & \mathcal{O} \text{ is 2-regular}, \\ \mathrm{Ind}_{H_1}^G(1), & \mathcal{O} \text{ is 2-regular}, \end{cases}$$

associated to the endoscopic H_i and the divisor $E_{\mathcal{O}}$.

Proof. The first part of the lemma is a special case of [H1, ch II, Proposition 9.1] (it also follows from our case-by-case calculations). The second part is verified by the calculations above. (For various facts about unipotent orbits, including the definition of induction used here, see [Spal].) □

§ 5 THE IGUSA EXPANSION FOR G_2

Notes on the derivation of the Igusa expansion. Consider an F-morphism

$$\phi : Y \to \mathbb{A},$$

smooth outside the "special fiber" $\phi^{-1}(0)$, from a smooth F-variety Y to the affine line over F. When we regard $Y(F)$ and $\mathbb{A}(F) = F$ as F-manifolds in the p-adic topology it will be explicitly stated. The "default" topology (to use the computer terminology) on Y is the Zariski topology. Let $f \in C_c^\infty(Y(F))$, considering $Y(F)$ as a p-adic manifold, and consider

$$F(\lambda) := \int_{\phi^{-1}(\lambda)(F)} f(y)|\omega_\lambda|,$$

for $\lambda \in F - \{0\}$. If f is locally a product of quasi-characters on a suitable local coordinate system of a neighborhoood of $\phi^{-1}(0)$ in the manifold $Y(F)$ then Igusa has given an "asymptotic expansion" for $F(\lambda)$ as $\lambda \to 0$. Actually, since the topology on $Y(F)$ is totally disconnected, this "asymptotic expansion" is actually an equality valid for all λ in a neighborhood of 0. We explain this next.

Assume that $Y(F)$ may be covered by patched U on which there exist coordinates $\mu_1, ..., \mu_n$ such that

(5.1) $\phi^{-1}(0)$ has normal crossings; more specifically, the irreducible components E_i of $\phi^{-1}(0) \cap U$ are of the form $\{\mu_i = 0\}$ on U and

$$\lambda = \alpha \prod_i \mu_i^{a_i},$$

for some integral $a_i > 0$, where α is a non-vanishing function in a p-adic neighborhood of 0 and $|\alpha| = q^{-m(\alpha)}$ is constant for all sufficiently small λ.

(5.2) On U, $f(y)$ is of the form

$$f(\mu_1, ..., \mu_n) = \gamma \prod_i \kappa_i(\mu_i),$$

where γ is constant for sufficiently small λ.

(5.3) the form ω is given on U by

$$\omega = \prod_i \mu_i^{b_i - 1} d\mu_1 ... d\mu_n.$$

(5.4) We **assume** that f vanishes off U and that if $b_i = 0$ then $\kappa_i \neq 1$, $1 \leq i \leq n$.

Let $F(\lambda)$ be given by (0.2) above. We will sketch the derivation of an equality of the form

$$F(\lambda) = \sum_{\theta,\beta} \theta(\lambda)|\lambda|^{\beta - 1} \sum_{r=1}^{e(\theta,\beta)} (-\log_q |\lambda|)^{r-1} F_r(\theta, \beta, f). \tag{5.5}$$

The following lemma helps us to compute F_1:

Lemma 5.6. *Let*

$$G(\lambda) = \sum_{\theta,\beta} \sum_{r=1}^{e(\theta,\beta)} a(\theta,\beta,r)(-\log_q|\lambda|)^{r-1}\theta(\lambda)|\lambda|^{\beta-1},$$

where the $a(\theta,\beta,r)$ are constants. If $a(\theta,\beta,r)=0$ for $r>1$ then the coefficients b_n in the power series of the truncated Mellin transform

$$M_\varepsilon G(s) := \int_{|\lambda|\le\varepsilon} G(\lambda)|\lambda|^s d\lambda = \sum_{n\ge 0} b_n q^{-ns}, \quad \varepsilon = q^{-M},$$

are given by

$$b_n = \begin{cases} vol(\mathcal{O}_F^\times, d\lambda)\sum_{\theta\ unramified,\beta} a(\theta,\beta,1)(\theta(\pi)q^{-\beta})^n, & n \ge M, \\ 0, & n < M. \end{cases}$$

The proof of this is left to the reader.

Now, to verify the formula (5.5) for $F(\lambda)$ we may verify it for its truncated Mellin transform and use the lemma above to check the coefficients of both sides of the resulting power series in q^{-s}. The formula we have in mind for F_1 is stated below. First, we need some notation: in the notation of the assumptions (5.1-4) above, let $c_j = c_j(a_1,...,a_n)$ denote the coefficients for which the principal part of $\prod_i \frac{1}{1-t^{a_i}}$ at $t=1$ is

$$\sum_j \frac{c_j}{(1-t)^j}.$$

Let

$$A(x) = \sum_j \frac{c_j}{(j-1)!}(x+1)...(x+j-1),$$

and define $A_i(x)$ by $A(x-y) = \sum_j x^{j-1}A_j(y)$, so $A_1(y)$ is the "constant term" of this polynomial in x. (The function $A(x)$ arises via the undergraduate calculus equality

$$\frac{(j-1)!}{(1-t)^j} = \sum_{n\ge 0}(n+1)(n+2)...(n+j-1)t^n.)$$

Lemma 5.7. *Suppose that $Y(F)$, f, ω, and λ satisfy the conditions of (5.1-4). Suppose furthermore that the support of f is contained in one such patch U, where*

$$U = \{(\mu_1,...,\mu_n) \in F^n \mid |\mu_i| \le q^{-M_i}\}.$$

Finally, suppose that we reindexed the μ_i, if necessary, so that

$$\theta^{a_i} = \kappa_i, \quad \beta = b_i/a_i, \quad 1 \le i \le p,$$

and for each $i > p$ at least one of these equalities fails to hold. (This p is of course different from the fixed prime p.) Then $F(\lambda)$ satisfies the formula (5.5) where $F_1(\theta,\beta,f)$ is given by the value of

$$\int_{D_p\cap U} A_1(M)h_p \prod_{i=p+1}^{n} |\mu_i|^{z_i}|\nu_p|,$$

at $z_{p+1} = ... = z_n = 0$, where

$$M = m(\alpha) + \sum_{i\le p} a_i M_i + \sum_{i>p} a_i(-\log_q|\mu_i|),$$

$$h_p = \frac{\gamma}{\theta(\alpha)} \prod_{i=p+1}^{n} \kappa_i(\mu_i)\theta(\mu_i)^{-a_i},$$

and

$$\nu_p = \frac{1}{|\alpha|^\beta} \prod_{i=p+1}^{n} |\mu_i|^{b_i - \beta a_i - 1} d\mu_{p+1} ... d\mu_n.$$

Moreover, if, in addition, any one of the characters $\kappa_i \theta^{-a_i}$ is ramified then $F_1(\theta, \beta, f) = 0$.

We now sketch the proof of this lemma following [L]. By our assumptions,

$$\begin{aligned} M_\varepsilon(F\theta^{-1})(s) &= \frac{|\alpha|^s}{\theta(\alpha)} \int_U f(y) \prod_i \theta(\mu_i)^{-a_i} |\mu_i|^{a_i s + b_i - 1} d\mu_1 ... d\mu_n \\ &= \frac{|\alpha|^s}{\theta(\alpha)} \gamma \prod_i \int_{|\mu| \leq q^{-M_i}} \kappa_i(\mu)\theta(\mu)^{-a_i} |\mu|^{a_i s + b_i - 1} d\mu, \end{aligned} \tag{5.8}$$

since $|\omega_\lambda||d\lambda| = \prod_{i=1}^n |\mu_i|^{b_i - 1} d\mu_1 ... d\mu_n$. Recall that

$$\int_{|\mu| \leq q^{-M_i}} \kappa'(\mu) |\mu|^z d\mu = 0$$

if κ' is a ramified character. Therefore we may assume from now on that every character $\kappa_i(\mu)\theta(\mu)^{-a_i}$ is unramified and we let

$$\varepsilon_i := \kappa_i(\pi)\theta(\pi)^{-a_i}, \quad 1 \leq i \leq n,$$

where π denotes a fixed generator of the maximal ideal of the local ring of integers $\mathscr{O}_F$. We may write

$$M_\varepsilon(F\theta^{-1})(s) = \gamma \frac{|\alpha|^s}{\theta(\alpha)} (1 - q^{-1})^n \prod_i \frac{(\varepsilon_i q^{-a_i s - b_i})^{M_i}}{1 - \varepsilon_i q^{-a_i s - b_i}}.$$

Recall that we have reindexed the μ_i so that

$$\theta^{a_i} = \kappa_i, \quad \beta = b_i / a_i, \quad 1 \leq i \leq p,$$

and for each $i > p$ at least one of these equalities fails to hold. Then

$$M_\varepsilon(F\theta^{-1})(s) = M(s, z_{p+1}, ..., z_n),$$

at $z_{p+1} = ... = z_n = 0$, where

$$
\begin{aligned}
&M(s, z_{p+1}, ..., z_n) \\
&\quad = (1-q^{-1})^p [\prod_{1\le i\le p} \frac{(q^{-a_i s - b_i})^{M_i}}{1-q^{-a_i s-b_i}}] |\alpha|^{s+\beta} \int_{\substack{|\mu_i|\le q^{-M_j} \\ p+1\le i\le n}} \times \\
&\qquad \times h_p|\alpha|^{-\beta} \prod_{p+1\le i\le n} |\mu_i|^{z_i + a_i s + b_i - 1} d\mu_{p+1} ... d\mu_n \\
&\quad = (1-q^{-1})^p |\alpha|^{s+\beta} \frac{\gamma}{\theta(\alpha)} [\prod_{1\le i\le p} \frac{(q^{-a_i s-b_i})^{M_i}}{1-q^{-a_i s-b_i}}] \times \\
&\qquad \times |\alpha|^{-\beta} \int_{\substack{|\mu_i|\le q^{-M_j} \\ p+1\le j\le n}} \prod_{p+1\le i\le n} |\mu_i|^{a_i(s+\beta)} \kappa_i(\mu_i) \theta(\mu_i)^{-a_i} |\mu_i|^{z_i - a_i\beta + b_i - 1} d\mu_{p+1} ... d\mu_n \\
&\quad = (1-q^{-1})^p |\alpha|^{s+\beta} [\prod_{1\le i\le p} \frac{(q^{-a_i s-b_i})^{M_i}}{1-q^{-a_i s-b_i}}] \times \\
&\qquad \times \int_{\substack{|\mu_i|\le q^{-M_j} \\ p+1\le i\le n}} \prod_{p+1\le i\le n} |\mu_i|^{z_i + a_i(s+\beta)} h_p|\nu_p|.
\end{aligned}
\tag{5.9}
$$

The last integral may be interpreted as a principal-value integral of a function on $Y(F)$ over a variety of the form $D_p = \cap_{1\le i\le p} E_i$, where E_i is an irreducible component of $\phi^{-1}(0)$ which satisfies $E_i \cap U = \{(\mu_1, ..., \mu_n) \mid \mu_i = 0\}$. Moreover, this multiple integral is a rational function of q^{-s}.

To see what this rational function is, let for the moment $t = q^{-s-\beta}$ and recall that we have defined the c_j in such a way that the principal part of

$$\prod_{1\le i\le p} \frac{(q^{-a_i(s+\beta)})^{M_i}}{1-q^{-a_i(s+\beta)}} = \prod_{1\le i\le p} \frac{t^{a_i M_i}}{1-t^{a_i}}$$

at $t = 1$ is

$$t^{\Sigma_{1\le i\le p} a_i M_i} \sum_{1\le i\le p} \frac{c_j}{(j-1)!} (1-t)^{-j} = \prod_{p+1\le i\le n} |\mu_i|^{a_i(s+\beta)} |\alpha|^{-s-\beta} t^M \sum_{n\ge 0} A(n) t^n,$$

where M is as in the statement of the lemma. (Here we have multiplied and divided by $\prod_{p+1\le i\le n} |\mu_i|^{a_i(s+\beta)} |\alpha|^{-s-\beta}$ to account for all of M.) The coefficient of t^n in the latter expression is, for $n \ge M$,

$$A(n-M) \prod_{p+1\le i\le n} |\mu_i|^{a_i(s+\beta)} |\alpha|^{-s-\beta} = \prod_{p+1\le i\le n} |\mu_i|^{a_i(s+\beta)} |\alpha|^{-s-\beta} \sum_r n^{r-1} A_r(M),$$

and, for $n < M$, the coefficient of t^n is 0. From this the lemma follows.

Geometry of the Igusa expansion. Let E be an irreducible divisor in $Y'' \subset G \times S''$, which we know is regular in codimension one, for which the image of E under π is contained in the Zariski closure of a unipotent orbit $\mathcal{O}$ in G. Such a divisor is called an **$\mathcal{O}$-divisor** if $\mathcal{O}$ is minimal (with respect to the partial ordering given by inclusion of Zariski closures) with this property. It will be called an **$\mathcal{O}(F)$-divisor** if E is defined over F and $\pi(E(F))$ meets $\mathcal{O}(F)$ in a dense subset of $\mathcal{O}(F)$ as p-adic manifolds. Let E be an $\mathcal{O}(F)$-divisor.

First, we record an observation about the fibration $\pi : E \to \overline{\mathcal{O}}$. As a corollary to [DG, ch II, Prop. 3.1], we have

Lemma 5.10. *(a) The morphism $\pi : E \cap Z_S \to \overline{\mathscr{O}}$ is smooth of relative dimension $2r$, provided $\mathscr{O}$ is r-regular.*

(b) $\pi(E \cap Z_S) \subset \mathscr{O}$.

(c) Let

$$\{1\} \subset \overline{\mathscr{O}}_1 = \mathscr{O}_1 \cup \{1\} \subset ... \subset \overline{\mathscr{O}} = \mathscr{O} \cup \mathscr{O}_m,$$

where each $\mathscr{O}_i$ is a unipotent orbit of G, and let $E_i = \pi^{-1}(\mathscr{O}_i) \cap E$. Then $\pi|_E : E_i \to \mathscr{O}_i$ is faithfully flat.

We may apply Igusa's expansion to

$$F(\lambda) = \int_{\phi^{-1}(\lambda)(F)} f(y)|\omega_\lambda|, \tag{5.11}$$

where f is any function on $Y_C(F)$ for which (a) f is compactly supported on the p-adic manifold $Y_C(F)$ and (b) ϕ, $|\omega_\lambda|$, Y_C, and f satisfy the conditions of Igusa data in [LS0]. Since f is compactly supported, so is $f|_{\mathscr{O}}$. On each fiber $\tilde{E}(u)$ the pull-back $\pi^*(f)$ takes a constant value $f(u)$. By [H1, Ch I], the regular function $\kappa(t_*)$ extends from $E(u)$ to a regular function on $\tilde{E}(u)$. By the lemma above, we have

$$\int_{\overline{\mathscr{O}}(F)} f \cdot PV \int_{\tilde{E}(u)(F)} \kappa(\tilde{t}_*)|\omega_{\tilde{E}(u)}||\omega_{\overline{\mathscr{O}}}| = PV \int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa(\tilde{t}_*)|\omega_{\tilde{E}}|, \tag{5.12}$$

provided everything in sight satisfies the conditions (5.1-4). Recall that, by definition, there are disjoint patches $U_i \subset \tilde{Y}_1(F)$, each analytically isomorphic to a finite box in affine space, such that

$$PV \int_{\tilde{E}(F)} \tilde{\pi}^*(f)\kappa(\tilde{t}_*)|\omega_{\tilde{E}}| = \sum_i PV \int_{\tilde{E}(F)\cap U_i} \tilde{\pi}^*(f)\kappa(\tilde{t}_*)|\omega_{\tilde{E}}|, \tag{5.13}$$

provided that the conditions (5.1-4) hold. (By the way, the computations in §4 show that in all the cases looked at in this paper, $b_{E(F)} > 0$. In particular, $b_{E(F)} \neq 0$.)

Lemma 5.14. *Let $G = G_2$. Let E be a base $\mathscr{O}(F)$-divisor of the variety $Y_C = \tilde{Y}_1$. If E is F-rational, $E(F) \neq \emptyset$, then $e(\theta_E, \beta_E) = 1$. In particular, spurious divisors do not contribute to the Igusa expansion.*

Proof. The proof is case-by-case using local coordinates in each case. Assume $e(\theta_E, \beta_E) \geq 2$. By [L], the principal value integral contributing to $F_1(\theta_E, \beta_E, f_\kappa)$ is

$$\int_{\mathscr{O}(F)} f \int_{D_u(F)} \kappa(\tilde{t}_*)|\omega_{D_u}||\omega_{\mathscr{O}}|,$$

where $D = \cap_{i=1}^{e(\theta_E,\beta_E)} E_i$, where $\theta_E = \theta_i$ and $\beta_E = \beta_i$, and D is F-rational (though the E_i themselves need not be F-rational).

Claim: The homogeneity degree of $|\omega_{D_u}|$, as a function of the tangent direction X, is $dim(D) - dim(\mathscr{O}) < dim(E) - dim(\mathscr{O}) = 2r$.

To prove this, we look at each divisor in the inverse image of $\{\lambda = 0\}$.

We may omit the regular and 1-regular cases since they were already dealt with in [H1], [LS2].

Let $\mathscr{O} = \mathscr{O}_2$ denote a 2-regular orbit and let $E = E_{\emptyset,\alpha,\delta}$. First suppose $e(\theta_E, \beta_E) = 2$. From §7 below, the form on the F-rational fiber $E_{\emptyset,\alpha,\delta}(u)$, $u \in \mathscr{O}$, is given on $Y''_{\emptyset,\alpha,\delta}$ by

$$\omega_{E_{\emptyset,\alpha,\delta}(u)} = g dz(\beta) dx(\zeta) dv(\alpha) dv(\zeta), \tag{5.15}$$

where g is a regular function. (Though these coordinates may not be F-rational, we may "average" them over the image of the Galois group in Ω using the calculations of §2 and then we make a

change of variables in the form to get a form in F-rational coordinates satisfying (5.1-4).) By (1.22), this form is homogeneous of degree 2 as a function of X. Let E' denote another F-rational $\mathscr{O}_2$-rational divisor with $\theta_E = \theta_{E'}$, $\beta_E = \beta_{E'}$. By Hironaka's theorem, E intersects E' normally in Y_C. The intersection $D = E \cap E'$ maps onto $\mathscr{O}$ under π by [DG, chap II, Prop 3.1]. The variety D must have codimension 1 in E and will have an Igusa form on a fiber $D(u)$ which satisfies

$$\omega_{D(u)} = g_D dx dv_1 dv_2,$$

where x is a coordinate on Z, v_1, v_2 are coordinates on the N_+ components, and g_D is regular. Therefore, $\omega_{D(u)}$ is homogeneous of degree < 2 as a function of X.

Now suppose $e(\theta_E, \beta_E) = 3$. The argument is similar, hence omitted, in this case. The claim follows in the 2-regular case.

Let $\mathscr{O} = \mathscr{O}_3$ denote a 3-regular orbit and let $E = E_{\emptyset,\beta,\zeta}$. First suppose $e(\theta_E, \beta_E) = 2$. From §8, the form on the F-rational fiber $E_{\emptyset,\beta,\zeta}(u)$, $u \in \mathscr{O}$, is given on $Y''_{\emptyset,\beta,\zeta}$ by

$$\omega_{E_{\emptyset,\beta,\zeta}(u)} = h dw(\varepsilon) dw(\zeta) dx(\zeta) dv(\beta) dv(\varepsilon) dv(\zeta), \tag{5.16}$$

where h is a regular function. (Again, these coordinates may be "averaged to obtain F-rational coordinates as in the 2-regular case.) By (1.22), this form is homogeneous of degree 3 as a function of X. Let E' denote another F-rational $\mathscr{O}_3$-rational divisor with $\theta_E = \theta_{E'}$, $\beta_E = \beta_{E'}$. By Hironaka's theorem, E intersects E' normally in Y_C. The intersection $D = E \cap E'$ maps onto $\mathscr{O}$ under π by [DG, chap II, Prop 3.1]. The variety D must have codimension 1 in E and will have Igusa form on a fiber $D(u)$ which satisfies

$$\omega_{D(u)} = h_D dx dy dv_1 dv_2 dv_3,$$

where x, y are coordinates on Z, v_1, v_2, v_3 are coordinates on the N_+ components, and h_D is regular. Therefore, $\omega_{D(u)}$ is homogeneous of degree < 3 as a function of X.

Now suppose $e(\theta_E, \beta_E) = 3$. The argument is similar, hence omitted, in this case. The claim follows in the 3-regular case. This proves the claim for all the orbits.

However, the claim contradicts the degree of homogeneity of the $\mathscr{O}$-germ. □

Lemma 5.17. *Let $G = G_2$. Let E be a base $\mathscr{O}(F)$-divisor of the variety Y_1, $\tilde{E}$ its preimage under $Y_C \to Y_1$. In this case $F_1(\theta_E, \beta_E, f_\kappa)$ is given (up to a constant) by*

$$PV \int_{\tilde{E}(F)} \tilde{\pi}^*(f_\kappa) |\omega_{\tilde{E}}| = \int_{\overline{\mathscr{O}}(F)_{st}} f \int_{\tilde{E}_u(F)} \kappa(\tilde{t}_*) |\omega_{\tilde{E}_u}| |\omega_{\overline{\mathscr{O}}}|,$$

where $\omega_{\overline{\mathscr{O}}_{st}}$ is a sum of forms of maximal degree on the equidimensional pieces $\mathscr{O}_i$ in Lemma 5.10 of the stable orbit $\overline{\mathscr{O}}(F)_{st} = \cup_i \mathscr{O}_i(F)$.

Proof. Because of the previous lemma, (0.1), and formula 0.6, it suffices to check that the conditions of [LS0] are met. This has essentially already been done in [L], [H1, ch I, proof of Prop. 5.2], but we fill in some details regarding the function $\kappa(\tilde{t}_*)$ and the cocycle-valued function t_*.

Some information about t_* is given following formula 0.6 in §0. More explicit information about $t_\sigma : Y_1(B_+, B_-) \to T_G(\overline{F})$, where $Y_1(B_+, B_-)$ is defined in §1, is given in [H1, ch I, Corollary 5.4]. Since the only F-form of G_2 over a p-adic field is the split form [T], we have $G = G_2 = G_{in} = G_{qs}$, $T_G = T_{in}$, $\underline{T} = T_{qs}$, and $\underline{B} = B_{qs}$, the notation being as in [H1]. Let $\alpha \in \Delta$ denote a simple root, If $\sigma = \sigma_\alpha \in \Gamma$ acts on $T_G(\overline{F})$ by conjugation by the reflection $s_\alpha \in \Omega$, then we choose g, h as in [H1, ch I, §5]. Taking $A_\sigma = w_\sigma^{-1} = s_\alpha \in N_G(T_G)$. in [H1, ch I, Corollary 5.7], we find there is a "constant" $t_1 \in T_G(\overline{F})$ independent of the star such that at a point

$$(t^g, ((B(W))_{W \in \mathscr{W}}, (z(\alpha))_{\alpha \in \Delta}, n) \in \chi^{-1}(C) \times S_1(B_+, B_-) \times N_+,$$

the value t_σ is determined by

$$s_\alpha \varepsilon_{-\alpha}(-\xi) n^{-1} w_\sigma^{-1} s_\alpha^{-1} \in N_+^- N_+ (t_1 t_\sigma)^{w_\sigma^{-1} s_\alpha^{-1}}. \tag{5.18}$$

Take the transpose of both sides of Equation 5.5 in [H1, ch I]. The resulting equation, along with (5.10) implies that

$$t_1 t_\sigma = \alpha^\vee(\xi),$$

where $\alpha^\vee$ is its coroot, and ξ satisfies (5.2) as a function of the coordinates of $Y''(B_+, B_-) \times N_+$. Choose B_-, B_+ such that these coordinates are F-rational. From the appendix, we know that there is a natural bijection

$$H^1(\Gamma, T_G) \to \mathbb{Z}/m\mathbb{Z},$$

where $m \in \{2,3\}$. Since each such $H^1(\Gamma, T_G)$ is cyclic, $\kappa(\tilde{t}_*)$ is determined by its value on a generator. This, in turn, determines a character of $\mathbb{Z}/m\mathbb{Z}$ and hence by local class field theory a character θ of $F^\times / N E_{T_G}^\times$ (where E_{T_G} denotes the Galois closure of the splitting field of T_G. One may construct a (non-canonical) map $\phi : H^1(\Gamma, T_G) \to F^\times$ such that $\kappa(\tilde{t}_*) = \theta(\phi(\tilde{t}_*))$ is a monomial in the coordinates of $Y''(B_+, B_-) \times N_+$. This allows us to conclude that the conditions of (5.1-4) have been met. □

We close this section by recalling an important reduction lemma due to T. Hales. First, observe that for each r-regular orbit $\mathcal{O}$, $r > 0$, every function $f \in C_c^\infty(G(F))$ may be written in the form $f = f_1 + f_2$ with f_1 supported off the Zariski closure of $\mathcal{O}$ in G and f_2 whose support does not meet the orbits outside the Zariski closure of $\mathcal{O}$ in G.

Reduction Lemma 5.19 (Hales). *Suppose that for each (T, κ) associated to H the germs $\Gamma_{\mathcal{O}}^{(T,\kappa)}(\gamma)$ have the form*

$$\Gamma_{\mathcal{O}}^{(T,\kappa)}(exp(\lambda^2 X)) = |\lambda|^{2r} m_{\mathcal{O}}^{(T,\kappa)}(X),$$

for λ small, where $m_{\mathcal{O}}^{(T,\kappa)}(X)$ satisfies

$$m_{\mathcal{O}}^{(T,\kappa)}(X) = m_{\mathcal{O}}^{(T,\kappa)}(X + Y),$$

for $X, Y \in \mathfrak{t}_G$ sufficiently small, X regular, $X+Y$ regular, and Y belonging to the Lie algebra of the center of an endoscopic group $H \subset G$ satisfying $Cent(Y, G) = H$. Then (G, H) admits local Δ-transfer at the identity for all functions $f \in C_c^\infty(G(F))$ whose support does not meet the orbits outside the Zariski closure of $\mathcal{O}$ in G.

Proof. Follows [H3, Lemma 6.1]. (See also [H2, Proposition 2.11].) □

This lemma implicitly uses [LS2].

§ 6 The matching for the 1-regular germ

Let $G = G_2$ throughout this section.

The κ-stable subregular germ for G_2 should match with the stable subregular germ of $PGL(3)$ or $(SL(2) \times SL(2))/\{\pm 1\}$, for homogeniety reasons. The idea to verify this is simply to write down the Igusa expansions of the two orbital integrals and match the $\beta = 2$ terms. Furthermore, for the 1-regular comparisons, we can and do assume that f is supported off the 0-regular orbit.

Let $u = exp(X_\alpha + X_{\alpha+\beta})$. Then the subregular orbit is given by $\mathcal{O}_1 = u^{G_2}$. This is not a stable orbit, but it is Richardson. See the appendix for more on its F-classes. It will turn out that both the biquadratic and the cubic Cartans can have non-zero 1-regular κ-stable germs.

From §§4,5, there are three base divisors with $\beta = 2$, and from [H1, ch. VI, §2], we know that all fundamental divisors of G_2 have $\theta = 1$. They are described as follows:

E_α is given locally as $\{\lambda = 0\}$ (with multiplicity one) on Z_α and Z_α is regular with generators of its local coordinate ring being λ, $x(\alpha)$, $z(\beta)$, $x(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$.

E_β is given locally as $\{\lambda = 0\}$ (with multiplicity one) on Z_β and Z_β is regular with generators of its local coordinate ring being λ, $z(\alpha)$, $x(\beta)$, $x(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$.

$E_{\emptyset,\alpha,\gamma}$ is given locally by $\{z(\alpha) = 0\}$ (with multiplicity one) on $Z_{\emptyset,\alpha,\gamma}$ and $Z_{\emptyset,\alpha,\gamma}$ is regular with generators of its local coordinate ring being $z(\alpha)$, $z(\beta)$, $x(\gamma)$, $w(\gamma)$, $x(\delta)$, $x(\varepsilon)$, $x(\zeta)$. This divisor may be eliminated from consideration by [H1, Ch V, Lemma 1.1].

As a corollary to [H1, ch 6, Lemma 1.1], we know that E_α and E_β are F-rational. Note that the action of the Galois group Γ on T_G and on Z factors through a finite group $\Gamma_T \subset \Omega$. To satisfy the hypotheses (5.1-4), one may use the calculations of §2 to "average" the coordinates of Z_α or Z_β over the Galois group Γ_T to get F-rational coordinates on these divisors.

Moreover, if we choose the Cartan T_G such that Γ acts on T_G via a product of two cyclic subgroups of order two of the Weyl group Ω then the κ-stable subregular germ of G_2 should match with the stable subregular germ of $(SL(2) \times SL(2))/\{\pm 1\}$ (and by the appendix, κ is quadratic) if we choose the Cartan T_G such that Γ acts on T_G via a cyclic subgroup of Ω of order three then the κ-stable subregular germ of G_2 should match with the stable subregular germ of $PGL(3)$ (and by the appendix, κ is a cubic character).

Theorem 6.1. *Suppose T_G is a non-split Cartan subgroup of G_2 which splits over a Galois extension for which the Galois group action on T_G is either a direct product of two cyclic subgroups of Ω order 2 or a cyclic subgroup of order 3. Assume that $f \in C_c^\infty(G(F))$ satisfies $supp(f) \cap \mathcal{O}_1 \neq \emptyset$, where $\mathcal{O}_1$ denotes a fixed 1-regular $G(F)$-orbit, and the support of f does not meet any unipotent orbit of $G(F)$ which is not 1-regular. Then there is an f^H, smooth on $H(F)$, for which, for all regular $\gamma_G \in T_G(F)$ in a sufficiently small neighborhood of 1, we have*

$$\Delta_{G/H}^{T_G,\kappa}(\gamma_H, \gamma_G)\Phi^{T_G,\kappa}(\gamma_G, f) = \Phi^{T_H,st}(\gamma_H, f^H),$$

where κ, T_G satisfy $H_{T_G,\kappa} = H_i$. Here $i = 1$, so $H = PGL(3)$, if the Galois group action on T_G is a cyclic subgroup of order 3, and $i = 2$, so $H = (SL(2) \times SL(2))/\{\pm 1\}$, if the Galois group action on T_G is a direct product of two cyclic subgroups of Ω order 2.

This will be proven below.

Proof of Theorem 6.1. By the work of Langlands and Shelstad, [LS1], [LS2], we need only prove the corresponding matching of germs conjecture stated in §0 above.

We discuss the Igusa expansion of the G_2 side more carefully: Since only the $\theta = 1$ term of formula 0.6(a) contributes by [H1, ch 6, §2], the $\beta = 2$ term of the Igusa expansion takes the form, $|\lambda|\kappa(inv(\gamma_G^0, \gamma_{G,\lambda}))D_H(\gamma_{H,\lambda})^{-1}$ times

$$F_1(1,2,f_\kappa) = PV \int_{U \cap E(F)} A_1(M)h_1|\nu_1| = PV \int_{E(F)\cap U} \pi^*(f)\kappa_1(t_*)|\omega_E| \tag{6.2}$$

where E is E_α or E_β, depending on T_G, and where $A_1(M)$, h_1, ν_1, ..., are as in [L], and f_κ includes the κ-factor. The factor $m_\kappa(e)/\kappa_1(\lambda)|_{E(u)}$ is given by [H1, Ch VI, §2 and Proposition 3.2], provided E is the divisor associated to a line of the Dynkin curve associated to the long root of G_2 which is fixed by the Galois action. By the homogeniety of germs, formula 0.6 and lemma 0.8, this factor is

equal to a multiple of

$$\sum_{\mathcal{O}\subset\mathcal{O}_1^{st}} \mu_{\mathcal{O}}(f)\Gamma_{\mathcal{O}}^{T_G,\kappa}(\gamma_\lambda),$$

where $\mathcal{O}_1^{st}$ denotes the stable 1-regular unipotent orbit. M. Assem has computed the number of F-rational classes in the stable subregular class when $p \neq 2,3$ (see the appendix below).

We discussed the geometry of the fibering $E_\alpha \to \overline{\mathcal{O}}_1$, where $\overline{\mathcal{O}}_1$ denotes the Zariski closure, in §5. We assume that the function f meets $\mathcal{O}_1^{st}$ and no other unipotent orbit. By formula 0.6 and (6.2), we have

$$PV \int_{U\cap E(F)} A_1(M)h_1|\nu_1| = \int_{\pi(U)\cap\mathcal{O}_1(F)} f \int_{E_u(F)} \kappa_1(t_*)|\omega_{E_\alpha(u)}||\omega_{\mathcal{O}_1}|, \tag{6.3}$$

where E denotes either E_α or E_β. We may restrict ourselves to the case where T_G is either biquadratic or cubic as in Theorem 6.1. The dependence of (6.3) on $C \subset exp(\overline{F}\cdot X)$, where $X \in \mathfrak{t}$ is a tangent vector for C, is via the dependence of $\kappa(t_*)$ and $\omega_{E_\alpha(u)}$ on X, where $\mathfrak{t}$ denotes the Lie algebra of T_G.

Recall [L, Lemma 2.12] that the measure ω_Y already incorporates the Weyl discriminant $\prod_{\sigma>0}(1-\sigma(\gamma)^{-1})$. Also, from §4, recall

$$|\omega_{Z_\alpha}| = \frac{\lambda}{z(\beta)^2} d\lambda dx(\alpha)dz(\beta)dx(\gamma)dx(\delta)dx(\varepsilon)dx(\zeta),$$

and

$$|\omega_{Z_\beta}| = \frac{\lambda}{z(\alpha)^2} d\lambda dz(\alpha)dx(\beta)dx(\gamma)dx(\delta)dx(\varepsilon)dx(\zeta).$$

Note that these coordinates are not all F-rational if c_σ is a simple reflection by (2.16), (2.17).

On $\mathcal{O}_1$ the roots "outside P_α" (or the roots "outside P_β") may be used to parameterize the orbit - see the lemma below.

The digression below is motivated by [H1, Remark V.2.2].

Let $\mathcal{O}$ be the Richardson orbit associated to a parabolic P as in [St, Theorem 2, §3.9]. Write $P = MN_P$, so that N_P^G contains $\mathcal{O}$ as a dense subset, $\mathcal{O}\cap N_P$ is dense in N_P and $\mathcal{O}\cap N_P$ forms a single class under P. We have dim$\mathcal{O}$ = dim$G$ - dim$M$. Let $u \in \mathcal{O}\cap N_P$ and recall $N_P^- P$ is dense in G by the Bruhat decomposition. The P-orbit $\{u^p \mid p \in P\}$ is equal to $\mathcal{O}\cap N_P$ and is dense in N_P. Therefore, the coordinates of N_P yield coordinates for the P-orbit of $u \in \mathcal{O}\cap N_P$. The G-orbit of this u, that is $\mathcal{O}$ itself, contains the $N_P^- P$-orbit of u as an open subvariety. We may regard the $N_P^- P$-orbit of u as the N_P^--orbit of $\mathcal{O}\cap N_P$. By Corollary 1.6, for each $\overline{n} \in N_P^-$, $B^{\overline{n}}$ is opposite to B^- and $B^{\overline{n}} \neq B$ unless $\overline{n} = 1$. Therefore, for our u as above, $u^{\overline{n}} \notin \mathcal{O}\cap N_P$. From this it follows that the N_P^--orbit of $\mathcal{O}\cap N_P$ is isomorphic to N_P^-.

Lemma 6.4 (Hales). *If $\mathcal{O}$ is the Richardson orbit associated to a parabolic P then the coordinates on $N_P^- \times N_P$ serve as local coordinates on $\mathcal{O}$.*

Returning to the proof of Theorem 6.1, associated to the roots outside P_α, we have the invariant measure

$$\omega_{\mathcal{O}_1} = \lim_{\lambda\to 0} \frac{dx(\beta)...dx(\zeta)dv(\beta)...dv(\zeta)}{\lambda},$$

Recall that the invariant measure $\omega_{\mathcal{O}_1}$ is unique up to a constant [R]. On the fiber $E_\beta(u)$, we use

$$\omega_{E_\beta(u)} = \frac{1}{z(\alpha)^2} dv(\alpha)dz(\alpha),$$

by §4 above, [L, Lemma 2.12]. We have then

$$\omega_{E_\beta(u)} = \frac{(1-\alpha(\gamma_\lambda)^{-1})}{\lambda}\Big|_{\lambda=0} dv(\alpha)\frac{dz_\alpha}{z_\alpha^2} = \alpha(X)dv(\alpha)\frac{dz_\alpha}{z_\alpha^2},$$

where $\alpha(X) := \frac{1-\alpha^{-1}(\gamma_\lambda)}{\lambda}|_{\lambda=0}$ denotes the inner product of α with the tangent vector X of C. Similarly, associated to the roots outside P_β, we have

$$\omega_{E_\alpha(u)} = \frac{1}{z(\beta)^2}dv(\beta)dz(\beta) = \beta(X)dv(\beta)\frac{dz_\beta}{z_\beta^2}.$$

Therefore, the $\beta = 2$ term is equal to

$$\begin{aligned} &|\lambda|\kappa(inv(\gamma_G^0,\gamma_{G,\lambda}))D_H(\gamma_{H,\lambda})^{-1}PV\int_{U\cap(E_\alpha)(F)} A_1(M)h_1|\nu_1| \\ &= |\lambda|\kappa(inv(\gamma_G^0,\gamma_{G,\lambda}))D_H(\gamma_{H,\lambda})^{-1}|\beta(X)|\int_{\mathscr{O}_1} f\int_{E_\alpha(u)\cap(U\cap\pi^{-1}(u))}\kappa(t_*)|dv'(\beta)\frac{dz_\beta}{z_\beta^2}||\omega_{\mathscr{O}_1}|, \end{aligned} \tag{6.5a}$$

or

$$\begin{aligned} &|\lambda|\kappa(inv(\gamma_G^0,\gamma_{G,\lambda}))D_H(\gamma_{H,\lambda})^{-1}PV\int_{U\cap(E_\beta)(F)} A_1(M)h_1|\nu_1| \\ &= |\alpha(X)|\int_{\mathscr{O}_1} f\int_{E_\beta(u)\cap(U\cap\pi^{-1}(u))}\kappa(t_*)|dv'(\alpha)\frac{dz_\alpha}{z_\alpha^2}||\omega_{\mathscr{O}_1}|, \end{aligned} \tag{6.5b}$$

From these expressions and the remarks in §0 above, one may verify the local translation invariance property of Hales' lemma 5.19. From this, the transfer follows in the 1-regular case, for compactly supported functions whose support meets the 1-regular orbit and no other orbit.

§ 7 On the 2-regular germ for G_2

Let $G = G_2$ throughtout this section.

The only essential base divisor which can contribute to the 2-regular germ for G_2 via Igusa theory is $E_{\emptyset,\alpha,\delta}$, which has $\beta(E_{\emptyset,\alpha,\delta}) = 3$. It must be F-rational since for every $\sigma \in \Gamma$ $\sigma(E_{\emptyset,\alpha,\delta})$ is also a 2-regular base divisor (see observation (I) in §4), so $\sigma(E_{\emptyset,\alpha,\delta}) = E_{\emptyset,\alpha,\delta}$. Recall that the action of the Galois group Γ on T_G and on Z factors through a finite group Γ_T, say. To satisfy the hypotheses (5.1-4), one may use the calculations of §2 to "average" the coordinates of $Z_{\emptyset,\alpha,\delta}$ over the Galois group Γ_T to get F-rational coordinates on this divisor if needed.

The branch $E_{\emptyset,\alpha,\delta}$ is defined by $\lambda = 0$ on the regular subset $Z_{\emptyset,\alpha,\delta}$: $\{x(\delta) \neq 0, w(\delta) \neq 0, z(\beta) \neq 0\} \subset Z$. The patch $Z_{\emptyset,\alpha,\delta}$ is regular with generators $z(\alpha)$, $z(\beta)$, $w(\gamma)$, $w(\delta)$, $x(\delta)$, $x(\varepsilon)$, and $x(\zeta)$. The divisor $\lambda = 0$ occurs on this patch as $z(\alpha) = 0$ with multiplicity 2. The local coordinate relations yield that on $E_{\emptyset,\alpha,\delta}$, we have $\lambda = x(\alpha) = x(\beta) = z(\alpha) = x(\gamma) = w(\varepsilon) = w(\zeta) = 0$.

By [H1, ch 6, §2], only the term $\theta = 1$ contributes to the sum over θ in formula 0.6(a). The contribution of the 2-regular germ to the κ-orbital integral is then $\kappa(inv(\gamma_G^0,\gamma_{G,\lambda}))D_H(\gamma_{H,\lambda})^{-1}$ times

$$|\lambda|^2 PV\int_{E_{\emptyset,\alpha,\delta}(F)} \pi^*(f)\kappa_1(t_*)|\omega_{E_{\emptyset,\alpha,\delta}}| = |\lambda|^2 PV\int_{\mathscr{O}_2} f\int_{E_{\emptyset,\alpha,\delta}(u)(F)}\kappa_1(t_*)|\omega_{E_{\emptyset,\alpha,\delta}(u)}|. \tag{7.1}$$

Theorem 7.2. *Suppose T_G is a non-split Cartan subgroup of G_2 which splits over a Galois extension for which the Galois group action on T_G is a direct product of two cyclic groups of order 2. Assume that $f \in C_c^\infty(G(F))$ satisfies $supp(f) \cap \mathscr{O}_2 \neq \emptyset$, where $\mathscr{O}_2$ is a fixed 2-regular $G(F)$-orbit, and the*

support of f does not meet unipotent orbit of $G(F)$ which is not 2-regular. Then there is an f^H, smooth on $H(F)$, for which, for all regular $\gamma_G \in T_G(F)$ in a sufficiently small neighborhood of 1, we have

$$\Delta_{G/H}^{T_G,\kappa}(\gamma_G)\Phi^{T_G,\kappa}(\gamma_G,f) = \Phi^{T_H,st}(\gamma_H,f^H),$$

where κ, T_G satisfy $H_{T_G,\kappa} = (SL(2)\times SL(2))/\{\pm 1\}$. (In particular, κ is quadratic and T_G is biquadratic.)

This will be proven below.

Proof of Theorem 7.2. By [Sh] or [Ro] the germ on the right-hand side is a non-zero constant. The theorem will follow provided we can show that both sides are equal, as a function of the tangent direction X, up to a constant factor. Indeed, such a constant factor can be absorbed into f^H provided we choose f in the beginning so that its support meets the 2-regular orbit and no other unipotent orbit. We fix such a choice for the remainder of this section.

We explicitly construct local coordinates for the unipotent orbit $\mathcal{O}_2^{st}$. Let $u = \theta_{\alpha+\beta}(1) \in G_2$, and compute the orbit $ad(\mathfrak{g}_2)(X_{\alpha+\beta})$. From this, we conclude that u^{G_2} is 8-dimensional, so $\mathcal{O}_2^{st} = u^{G_2}$. Analogously to Hales' lemma 6.4, we find that the coordinates of the subvariety $\prod_{\sigma\in\Sigma_G-\Sigma_{H_2}} U_\sigma$ of G_2 yields local coordinates for the 2-regular orbit. Here $\Sigma_{H_2}^+ = \{\alpha,\zeta\}$. There is only one F-orbit of u contained in the stable orbit of u (see the appendix), so the 2-regular orbit of G_2 is stable. Local coordinates for the fiber $E_{\emptyset,\alpha,\delta}(u)$ may be choosen from the remaining coordinates on $E_{\emptyset,\alpha,\delta}$: $z(\beta)$, $x(\zeta)$, $v(\alpha)$, and $v(\zeta)$.

We use the form on $\mathcal{O}_2$,

$$\begin{aligned}\omega_{\mathcal{O}_2} &:= \frac{|\bigwedge_{\sigma\in\Sigma_{\mathcal{O}_2}} dx(\sigma)\wedge dv(\sigma)|}{|\lambda|^{3/2}} \\ &= \frac{|x(\delta)|^{1/2}}{|w(\delta)|^{3/2}|z(\beta)|^{3/2}} dz(\alpha)dw(\gamma)dw(\delta)dv(\alpha)dv(\gamma)dv(\delta),\end{aligned} \tag{7.3}$$

where $\Sigma_{\mathcal{O}_3} = \Sigma_G^+ - \Sigma_{H_2}^+$. On $E_{\emptyset,\alpha,\delta}(u)$,

$$\omega_{E_{\emptyset,\alpha,\delta}(u)} := \frac{|x(\delta)|^{5/2}|w(\delta)|^{7/2}}{|z(\beta)|^{1/2}} dz(\beta)dx(\zeta)dv(\alpha)dv(\zeta). \tag{7.4}$$

We need to rewrite this last form as an expression

$$\omega_{E_{\emptyset,\alpha,\delta}(u)} = |p_{\mathcal{O}_2,T_G}(X)|\omega'_{E_{\emptyset,\alpha,\delta}(u)},$$

where $p_{\mathcal{O}_2,T_G}(X)$ is a homogeneous polynomial in the coordinates of tangent vector X (expressed in a basis of the Lie algebra $\mathfrak{t}$ of T_G) and $\omega'_{E_{\emptyset,\alpha,\delta}(u)}$ is independent of the tangent direction. Using (1.22), we find that $p_{\mathcal{O}_2,T_G}(X)$ is a product of the form

$$p_{\mathcal{O}_2,T_G}(X) = \prod_{\sigma\in\Sigma_{H_2}^+} \sigma(X)^{f_\sigma}, \tag{7.5}$$

where (by homogeneity of germs) $\sum_{\sigma>0} f_\sigma = 2$. To verify the hypotheses to Hales' lemma 5.19 one must verify the translation invariance property of (7.1). The translation invariance of this factor (7.5) follows from the above expression for $|p_{\mathcal{O}_2,T_G}(X)|$ as a product over roots in H_2.

By continuity, $\kappa(t_*)$ satisfies the local translation invariance property as well for sufficiently small Y. Therefore, the hypotheses to Hales' reduction lemma hold and the theorem follows in this case from lemma 5.19.

§ 8 On the 3-regular germ for G_2

Let $G = G_2$ throughtout this section.

The only essential base divisor which can contribute to the 3-regular germ for G_2 via Igusa theory is $E_{\emptyset,\beta,\zeta}$, which has $\beta(E_{\emptyset,\beta,\zeta}) = 4$. It must be F-rational since $\sigma(E_{\emptyset,\beta,\zeta})$ is also a 3-regular base divisor, by the observation (I) of §4 and the case $G = G_2$ of conjecture 3.4 (which we proved in §3 above). Recall that the action of the Galois group Γ on T_G and on Z factors through a finite group Γ_T. To satisfy the hypotheses (5.1-4), one may use the calculations of §2 to "average" the coordinates of $Z_{\emptyset,\beta,\zeta}$ (or a suitable $\mathbb{Z}$-linear combination of them) over Γ_T to get F-rational coordinates on this divisor.

The branch $E_{\emptyset,\beta,\zeta}$ is defined by $\lambda = 0$ on the regular subset $Z_{\emptyset,\beta,\zeta}$: $\{x(\zeta) \neq 0, w(\zeta) \neq 0, z(\alpha) \neq 0\} \subset Z$. If we use the local coordinates $z(\alpha)$, $z(\beta)$, $w(\gamma)$, $w(\delta)$, $w(\varepsilon)$, $w(\zeta)$, and $x(\zeta)$ on $Z_{\emptyset,\beta,\zeta}$ to compute the form ω_E on $E = E_{\emptyset,\beta,\zeta}$, we get

$$\begin{aligned}\omega_{E_{\emptyset,\beta,\zeta}} &= \frac{\omega_{Z_1}}{\lambda^{\beta}\frac{dz(\beta)}{z(\beta)}} \\ &= \frac{d\lambda \wedge dx(\alpha) \wedge \ldots \wedge dx(\zeta)}{\lambda^4 \frac{dz(\beta)}{z(\beta)}} \\ &= \frac{x(\zeta)^2 w(\zeta)^4}{w(\delta)^7 z(\alpha)^2} dz(\alpha) \wedge dw(\gamma) \wedge \ldots \wedge dw(\zeta) \wedge dx(\zeta).\end{aligned} \tag{8.1}$$

The divisor $\lambda = 0$ occurs on $Z_{\emptyset,\beta,\zeta}$ as $z(\beta) = 0$ with multiplicity 2. On $E_{\emptyset,\beta,\zeta}$, $\lambda = z(\beta) = x(\varepsilon) = x(\delta) = x(\gamma) = x(\beta) = x(\alpha) = 0$.

By §§4-5 and formula 0.6, the contribution of the 3-regular germ to the κ-orbital integral is therefore, $\kappa(inv(\gamma_G^0, \gamma_{G,\lambda}))D_H(\gamma_{H,\lambda})^{-1}$ times

$$|\lambda|^3 \sum_{\theta} \theta(\lambda) F_1(\theta, 4, \pi^*(f)\kappa_\theta(t_*)) = |\lambda|^3 \sum_{\theta} \theta(\lambda) \int_{E_{\emptyset,\beta,\zeta}(F)} \pi^*(f)\kappa_\theta(t_*)|\omega_{E_{\emptyset,\beta,\zeta}}|.$$

Note that only $\theta = 1$ contributes to the sum by [H1, ch 6, §2]. Recall that, on $Z_{\emptyset,\beta,\zeta}$,

$$|\omega_{E_{\emptyset,\beta,\zeta}}| = \frac{|x(\zeta)|^6 |z(\alpha)|^{10}}{|w(\delta)|^7} dz(\alpha) dw(\gamma) dw(\delta) dw(\varepsilon) dw(\zeta) dx(\zeta) dv(\alpha) \ldots dv(\zeta).$$

We may assume that the support of f meets $\mathcal{O}_3^{st}$ and no other unipotent orbit. In this case, by Formula 0.6(b), we have

$$\int_{E_{\emptyset,\beta,\zeta}(F)} \pi^*(f)\kappa_\theta(t_*)|\omega_{E_{\emptyset,\beta,\zeta}}| = \int_{\mathcal{O}_3(F)} f \int_{E_{\emptyset,\beta,\zeta}(u)} \kappa_1(t_*)|\omega_{E_{\emptyset,\beta,\zeta}(u)}||\omega_{\mathcal{O}_3}|, \tag{8.2}$$

which is a multiple of the κ-stable unipotent orbital integral of f associated to the $3-$regular orbit of G_2. It is shown in the appendix that the 3-regular orbit is stable.

Theorem 8.3. *Suppose T_G is a non-split Cartan subgroup of G_2 which splits over a Galois extension for which the Galois group action on T_G is realized by a cyclic subgroup of Ω of order three. Assume that $f \in C_c^\infty(G(F))$ satisfies $supp(f) \cap \mathcal{O}_3 \neq \emptyset$ and the support of f does not meet other unipotent orbit of $G(F)$. Then there is an f^H, smooth on $H(F)$, for which, for all regular $\gamma_G \in T_G(F)$ in a sufficiently small neighborhood of 1, we have*

$$\Delta_{G/H}^{T_G,\kappa}(\gamma_H, \gamma_G)\Phi^{T_G,\kappa}(\gamma_G, f) = \Phi^{T_H,st}(\gamma_H, f^H),$$

where κ, T_G satisfy $H_{T_G,\kappa} = PGL(3)$. (In particular, κ is a cubic character and T_G is a cyclic "cubic" elliptic Cartan.)

This will be proven below.

Proof of Theorem 8.3. Note that by [Ro], the germ on the right-hand side, $\Gamma_{\mathcal{O}_H^{st}}^{T_H,st}(\gamma_H)$, is a non-zero constant. Therefore, the theorem will follow if we can show that, as a function of the tangent vector X of C, the two sides are equal up to a constant.

Let $u = \theta_\beta(1) \in G_2$. One may explicitly compute the orbit $ad(\mathfrak{g}_2)(X_\beta)$. From this, we conclude that u^{G_2} is 6-dimensional. Therefore $\mathcal{O}_3^{st} = u^{G_2}$. There is only one F-orbit of u contained in the stable orbit of u (as is verified in the appendix), so the 3-regular orbit of G_2 is stable.

To get the local coordinates on the fiber $E_{\emptyset,\beta,\zeta}(u) \subset E_{\emptyset,\beta,\zeta}$ over u in the 3-regular orbit $\mathcal{O}_3$ take the local coordinates in the "right" divisor $E_{\emptyset,\beta,\zeta}$, then divide $\omega_{E_{\emptyset,\beta,\zeta}}$ by a Haar measure $\omega_{\mathcal{O}_3}$ on $\mathcal{O}_3$. Analogously to Hales' lemma 6.4, the subvariety $\prod_{\sigma\in\Sigma_G-\Sigma_{H_1}} U_\sigma$ of G_2 yields local coordinates for the 3-regular orbit, where $\Sigma_{H_1}^+ = \{\beta,\varepsilon,\zeta\}$. We use the form on $\mathcal{O}_3$,

$$\begin{aligned} |\omega_{\mathcal{O}_3}| &:= \frac{|\bigwedge_{\sigma\in\Sigma_{\mathcal{O}_3}} dx(\sigma)\wedge dv(\sigma)|}{|\lambda|^2} \\ &= \frac{|x(\zeta)|}{|w(\zeta)||z(\alpha)|} dz(\alpha)dw(\gamma)dw(\delta)dv(\alpha)dv(\gamma)dv(\delta), \end{aligned} \tag{8.4}$$

where $\Sigma_{\mathcal{O}_3} = \Sigma_G^+ - \Sigma_{H_1}^+$. On the generic fiber $E_{\emptyset,\beta,\zeta}(u)$, $u \in \mathcal{O}_3$,

$$|\omega_{E_{\emptyset,\beta,\zeta}(u)}| = \frac{|x(\zeta)||w(\zeta)|^5}{|z(\alpha)||w(\delta)|^7} dw(\varepsilon)dw(\zeta)dx(\zeta)dv(\beta)dv(\varepsilon)dv(\zeta). \tag{8.5}$$

We need to rewrite this last form as an expression

$$\omega_{E_{\emptyset,\beta,\zeta}(u)} = |p_{\mathcal{O}_3,T_G}(X)|\omega'_{E_{\emptyset,\beta,\zeta}(u)},$$

where $p_{\mathcal{O}_3,T_G}(X)$ is a homogeneous polynomial in the coordinates of tangent vector X to be computed below and $\omega'_{E_{\emptyset,\beta,\zeta}(u)}$ is independent of the tangent direction.

By an easy calculation given in the appendix, our assumption on T_G implies that κ is a character of order 3.

Once we do this it will follow that

$$\begin{aligned} &D_H(\gamma_{\lambda,H}) \sum_{\mathcal{O}\subset\mathcal{O}_3^{st}} \mu_{\mathcal{O}}(f)\Gamma_{\mathcal{O}}^{T_G,\kappa}(\gamma_\lambda) = |\lambda|^2\kappa(inv(\gamma_G^0,\gamma_\lambda))\int_{E_{\emptyset,\beta,\zeta}} \pi^*(f)\kappa_1(t_*)|\omega_{\mathcal{E}_{\emptyset,\beta,\zeta}(F)}| \\ &= |\lambda|^2\kappa(inv(\gamma_G^0,\gamma_\lambda))|p_{\mathcal{O}_3,T_G}(X)| \sum_{\mathcal{O}\subset\mathcal{O}_3^{st}} \int_{\mathcal{O}_3(F)} f \int_{E_{\emptyset,\beta,\zeta}(u)(F)} \kappa(t_*)|\omega'_{E_{\emptyset,\beta,\zeta}(u)}||\omega_{\mathcal{O}}|, \end{aligned} \tag{8.6}$$

where we normalize $\omega_{\mathcal{O}}$ consistent with the Langlands-Shelstad transfer factor as in [H2].

To determine the general form of $p_{\mathcal{O}_3,T_G}(X)$ we use the coordinate relations in (1.22). We find that $p_{\mathcal{O}_3,T_G}(X)$ is a product of the form

$$p_{\mathcal{O}_3,T_G}(X) = \prod_{\sigma\in\Sigma_H^+} \sigma(X)^{e_\sigma}, \tag{8.7}$$

where $\sum_{\sigma\in\Sigma_H^+} e_\sigma = 3$. One interesting thing to note is that the product runs over roots "in H".

To complete the matching, it remains only to verify the local translation invariance property in the hypothesis to Hales' lemma 5.19 which allows us to apply the descent argument. By continuity,

we see that the factor $\kappa(inv(\gamma_G^0, \gamma_{G,\lambda}))\kappa(t_*)$ satisfies the local translation invariance property of lemma 5.19 above, for sufficiently small Y. One checks that the factor (8.7) satisfies the translation invariance property using that it is a product over roots in H. The theorem now follows from lemma 5.19.

§ 9 A CONJECTURE

We conclude with a conjecture, which seems to be indicated by the proof above.

Conjecture 9.1. *Suppose T_G is an elliptic Cartan subgroup of a connected, simply connected, F-quasi-split group G. Let $\mathcal{O}$ denote an r-regular unipotent orbit of $G(F)$, let κ denote a character of $H^1(\Gamma, T_G)$, and let $H = H_{\kappa,T_G}$ denote an endoscopic group of G [L′]. Assume*

(1) there is an essential, admissible F-rational $\mathcal{O}$-divisor $E = E_{S_1,S_2,\sigma}$ such that, for each $u \in \mathcal{O}$, the Igusa form on the fiber satisfies

$$\omega_{E(u)} = p_{\mathcal{O},T_G}(X)\omega'_{E(u)},$$

where $\omega'_{E(u)}$ is independent of X and where

$$p_{\mathcal{O},T_G}(X) = \prod_{\sigma \in \Sigma^+_{\mathcal{O}}} \sigma(X),$$

for some $\Sigma^+_{\mathcal{O}} \subset \Sigma^+_G$,

(2) the function $p_{\mathcal{O},T_G}$ in (1) satisfies the hypotheses to Hales' reduction lemma,

(3) $b_{E,i} \neq 0$ in the F-rational coordinates of $Y''(B_+)$,

(4) $e(\theta, r+1) \leq 1$ and $e(\theta, r+1) = 0$ unless $\theta = 1$,

(5) there is only one r-regular unipotent orbit $\mathcal{O}$ of $G(\overline{F})$, and it is either Richardson or endoscopically Richardson,

(6) there is only one r-regular orbit $\mathcal{O}_H$ of $H(\overline{F})$.

Then the matching of germs conjecture stated in the introduction holds between these r-regular orbits. Moreover, for each $f \in C_c^\infty(G(F))$ such that $supp(f) \cap \mathcal{O}' = \emptyset$ if $\mathcal{O}'$ is not stably conjugate to $\mathcal{O}$, we have, for all regular $\gamma_G \in T_G(F)$ in a sufficiently small neighborhood of 1 and for any function f,

$$\Delta^{T_G,\kappa}_{G/H}(\gamma_H, \gamma_G)\Phi^{T_G,\kappa}(\gamma_G, f) = \Phi^{T_H,st}(\gamma_H, f^H),$$

for some $f^H \in C_c^\infty(H(F))$ such that $supp(f^H) \cap \mathcal{O}'_H = \emptyset$ if $\mathcal{O}'_H$ is not stably conjugate in H to $\mathcal{O}_H$.

Finally, we remark that there are a several low-dimensional cases which appear to satisfy the hypotheses of this conjecture. We mention one:

$G = SL(4)$, $\mathcal{O} = \mathcal{O}_{(2,2)}$ the 2-regular orbit associated to the partition $4 = 2+2$, $H = GL(2,E)^1$ where E/F is quadratic and the superscript one denotes the kernel of the map $Nm_{E/F} \circ \det : GL(2,E) \to F^\times$, and $\mathcal{O}_H$ is the orbit of $H(F)$ associated to the trivial 1-regular orbit of $GL(2,E)$ by restriction of scalars.

APPENDIX: BACKGROUND ON G_2

The first four sections of this appendix have been taken from an unpublished joint paper with D. Bump.

The purpose of this appendix is to collect some foundational material on the exceptional group G_2, then to use this material to compute the cuspidal endoscopic groups of G_2, the groups $H^1(\Gamma, T)$, and the F-classes in the stable 2- and 3-regular unipotent orbit.

In the first few sections, we shall be concerned with the phenomenon known as "triality", which yields an embedding of G_2 into the double cover Spin(8) of the split form of SO(8). We shall determine the triality map explicitly at the root level, and also use it to give an explicit Chevalley basis for the Lie algebra $\mathfrak{g}_2$, which should be useful in other calculations.
Acknowledgement: We thank Becci Davies for coding the root diagrams (in AMS–TEX).

Background on G_2 and SO(8).

A.1. SO(8). Let J denote the 8×8 matrix with 1's on the skew–diagonal and 0's elsewhere:

$$J := \begin{pmatrix} 0 & & 1 \\ & \iddots & \\ 1 & & 0 \end{pmatrix}.$$

Define the 8×8 **special orthogonal group** by

$$SO(8) := \{A \in GL(8) \mid A \cdot J \cdot {}^tA = J\}^o,$$

so that $SO(8)$ is semi–simple, connected, neither simply connected nor adjoint, non–compact, and split over $\mathbb{Q}$. Its maximal $\mathbb{Q}$-split torus consists of all matrices of the form

$$\text{(A.1)} \qquad \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \text{ with } A := \begin{pmatrix} a_1 & & & \\ & a_2 & & \\ & & a_3 & \\ & & & a_4 \end{pmatrix}, \; B := \begin{pmatrix} a_4^{-1} & & & \\ & a_3^{-1} & & \\ & & a_2^{-1} & \\ & & & a_1^{-1} \end{pmatrix}.$$

The compact real form of $SO(8)$ is obtained, of course, by replacing J by the 8×8 identity matrix.

The simply–connected covering group of $SO(8)$ is $SO(8)_{sc} =$ Spin(8), which has center $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and since $SO(8)$ has center $\mathbb{Z}/2\mathbb{Z}$, Spin(8) is a 2-fold cover of $SO(8)$ and a 4-fold cover of $PSO(8)$. The outer automorphisms of Spin(8) are in one–to–one correspondence with the automorphisms of the Dynkin diagram for $SO(8)$:



The Lie algebra

$$so(8) := \{A \in M_8 \mid AJ + J^tA = 0\}$$

may be expressed as the Lie algebra of 8×8 matrices of the form

$$\text{(A.2)} \qquad A = \begin{pmatrix} t_1 & a_{12} & a_{13} & a_{14} & a_{15} & a_{16} & a_{17} & 0 \\ a_{21} & t_2 & a_{23} & a_{24} & a_{25} & a_{26} & 0 & -a_{17} \\ a_{31} & a_{32} & t_3 & a_{34} & a_{35} & 0 & -a_{26} & -a_{16} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & -a_{71} & -a_{61} & -a_{51} & -a_{41} & -a_{31} & -a_{21} & -t_1 \end{pmatrix}.$$

If $i \neq j$ then define E_{ij} to be the matrix A as above with $a_{ij} = 1$ and all other entries = 0. Define E_{ii} to be the matrix A as above with $t_i = 1$ and all other entries = 0. Then

$$\begin{aligned}&\{E_{11}, E_{22}, E_{33}, E_{44},\\ &E_{12}, E_{13}, \ldots, E_{17}, E_{23}, \ldots, E_{26}, E_{34}, E_{35},\\ &{}^tE_{12}, {}^tE_{13}, \ldots, {}^tE_{17}, {}^tE_{23}, \ldots, {}^tE_{26}, {}^tE_{34}, {}^tE_{35}\}\end{aligned} \tag{A.3}$$

forms a basis for the 28-dimensional vector space $so(8)$.

A.2. The Cayley algebra and G_2. The **split Cayley algebra** is the 2×2 matrix algebra

$$C := \left\{ \begin{pmatrix} \alpha & a \\ b & \beta \end{pmatrix} \mid \alpha, \beta \in \mathbb{Q},\ a, b \in V \right\}, \tag{A.4}$$

where $V \cong \mathbb{Q}^3$ is the cross product algebra

$$V := \{xi + yj + zk \mid x, y, z \in \mathbb{Q},\ i \times i = j \times j = k \times k = 0,\\ i \times j = k,\ j \times k = i,\ k \times j = i,\ j \times i = -k, \text{ etc.}\}.$$

Define

$$G_2(F) := \mathrm{Aut}_F(C \otimes F), \tag{A.5}$$

where F is any field containing $\mathbb{Q}$. Since Der(C) $\cong$ Lie(Aut C), it follows that the Lie algebra $\mathfrak{g}_2$ may be identified with Der(C). For G_2, there are 2 real forms: one being the non-compact $\mathbb{Q}$–split from above, the other being a compact form associated to a division algebra (see[Jac, pp. 142-147] or [T]). We have

$$\dim_{\mathbb{Q}} C = 8,\ \dim_{\mathbb{Q}} \mathfrak{g}_2 = 14,$$

and there is a trace bilinear form which induces an embedding

$$G_2 = \mathrm{Aut}(C) \hookrightarrow SO(8).$$

Here SO(8) is regarded as the special orthogonal group associated to the trace form $trace(\beta)$ on C, where

$$\beta(x, y) = \frac{1}{2}(x \cdot y^* + y^* \cdot x), \quad \beta(x, x) = N(x),$$

and where N denotes the norm on C. Regarding G_2 as a subgroup of $SO(8)$, we have the following

Triality theorem A.6([Jac])**.** *Given any orthogonal map*

$$A_0 : C \to C,$$

there are orthogonal maps $A_1,\ A_2$ such that, for all $x, y \in C$,

$$A_0(xy) = A_1(x)A_2(y).$$

Moreover, the only other orthogonal maps satisfying this property are $-A_1,\ -A_2$, so

$$A_0 \longmapsto (A_1, A_2)$$

is well–defined, up to signs.

Proof. We will verify that our form of triality is the global equivalent of Jacobson's form [Jac, p. 8]. Let

$$A_0 = 1 + \delta B_0,$$
$$A_1 = 1 + \delta B_1,$$
$$A_2 = 1 + \delta B_2,$$

in $SO(8,\mathbb{Q})$ (regarded as a group of orthogonal maps with respect to the trace form on C), where $\delta^2 = 0$ and $B_i \in so(8,\mathbb{Q})$. Thus $A_0,\ A_1,\ A_2$ may be regarded as lying in the tangent space to SO(8) at the identity (cf. [Bor, chapter 0]). Then $A_0(xy) = A_1(x)A_2(y)$ is equivalent to

$$B_0(xy) = B_1(x)y + xB_2(y).$$

This is Jacobson's expression. □

Corollary A.7. *The map*

$$\begin{aligned} SO(8,F) &\to PSO(8,F), \\ A_0 &\longmapsto \quad \pm A_1, \end{aligned}$$

defines a projective representation of SO(8), i.e., belongs to PSO(8). In particular, it induces an outer automorphism of degree 3 (the "triality" automorphism) corresponding to a rotation of the Dynkin diagram of $PSO(8,F)$.

Remarks (A.8) If $A_0 \in G_2 = \mathrm{Aut}(C)$ then triality sends $A_0 \longmapsto A_0$. Conversely, if triality sends $A_0 \longmapsto A_0$ then $A_0 \in G_2$, so G_2 is the subgroup fixed by triality. In general, triality sends (up to signs)

$$A_0 \longmapsto A_1,\ A_1 \longmapsto A_2,\ A_2 \longmapsto A_0.$$

This way we can explicitly construct an outer automorphism of $SO(8)$.

(A.9) Triality can be described explicitly on the standard Borel subgroup B of PSO(8): triality cyclically permutes the "root groups" U_α, defined below, in the same way as it permutes the simple roots via its action on the Dynkin diagram; it also permutes the embeddings $PSL(2) \hookrightarrow PSO(8)$ associated to the simple roots. The images of the maximal split torus $T \subset PSL(2)$ embedded in $PSO(8)$ generate the maximal split torus in $PSO(8)$ and, of course, the simple root groups generate the unipotent radical of B. Putting these all together explicitly gives the action of triality on B.

A.3. Root systems for G_2 and $SO(8)$. Since the symmetric group S_3 acts on the Dynkin diagram of $SO(8)$ by triality, it acts on its 24 roots. Of these, 6 roots remain fixed under this action, and 18 are permuted cyclically in groups of three. In particular, the roots corresponding to

$$\{E_{17}\},\ \{E_{16}\},\ \{E_{23}\},$$

are fixed and the triples of roots corresponding to

$$\{E_{14},\ E_{15},\ E_{26}\},\ \{E_{13},\ E_{24},\ E_{25}\},\ \{E_{12},\ E_{34},\ E_{35}\},$$

are each permuted by the action of S_3. More explicitly, let us represent a basis of $so(8)$ by

$$\varepsilon_1 - \varepsilon_2,\ \varepsilon_2 - \varepsilon_3,\ \varepsilon_3 - \varepsilon_4,\ \varepsilon_3 + \varepsilon_4,$$

where

$$\varepsilon_i : \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \longmapsto a_i, \tag{A.10}$$

in the notation of (A.1), $1 \le i \le 4$. Here E_{12} corresponds to $\varepsilon_1 - \varepsilon_2$, E_{13} to $\varepsilon_1 - \varepsilon_2 + \varepsilon_2 - \varepsilon_3 = \varepsilon_1 - \varepsilon_3$, etc. Since $\varepsilon_2 - \varepsilon_3$ is connected to the others (since its Killing form inner product with them is

non–zero), it corresponds to the center of the Dynkin diagram

$$\begin{array}{ccc} & & \bullet\ \varepsilon_1-\varepsilon_2 \\ \varepsilon_3-\varepsilon_4\ \bullet & & \bullet\ \varepsilon_2-\varepsilon_3 \\ & & \bullet\ \varepsilon_3+\varepsilon_4 \end{array}$$

Therefore, under the action of S_3, the triality automorphism of so(8) induces a linear map ϕ on the roots such that

(A.11) $\phi(\varepsilon_1-\varepsilon_2)=\varepsilon_3-\varepsilon_4,\ \phi(\varepsilon_3-\varepsilon_4)=\varepsilon_3+\varepsilon_4,\ \phi(\varepsilon_3+\varepsilon_4)=\varepsilon_1-\varepsilon_2,\ \phi(\varepsilon_2-\varepsilon_3)=\varepsilon_2-\varepsilon_3.$

Thus

$$\begin{aligned}\phi(\varepsilon_1)&=\frac{1}{2}(\varepsilon_1+\varepsilon_2+\varepsilon_3-\varepsilon_4),\\ \phi(\varepsilon_2)&=\frac{1}{2}(\varepsilon_1+\varepsilon_2-\varepsilon_3+\varepsilon_4),\\ \phi(\varepsilon_3)&=\frac{1}{2}(\varepsilon_1-\varepsilon_2+\varepsilon_3+\varepsilon_4),\\ \phi(\varepsilon_4)&=\frac{1}{2}(\varepsilon_1-\varepsilon_2-\varepsilon_3-\varepsilon_4).\end{aligned}$$

Call the induced automorphism on the Lie algebra so(8), ϕ^*. As we've already observed, the Lie subalgebra

$$\{X\in so(8)\mid \phi^*(X)=X\}$$

is the Lie algebra $\mathfrak{g}_2$. Identifying its Lie group with a subgroup of $SO(8)$, we find that its maximal $\mathbb{Q}-$split torus is of the form

(A.12)
$$T':=\left\{\begin{pmatrix}A&0\\0&B\end{pmatrix},\ |\ A:=\begin{pmatrix}a_1&&\\&a_2&\\&&a_3\end{pmatrix},\ B:=\begin{pmatrix}1&&\\&a_3^{-1}&\\&&a_2^{-1}\end{pmatrix}\right.$$
$$\left.\begin{pmatrix}&&\\&&\\&&a_1^{-1}\end{pmatrix}\ \text{and}\ a_1=a_2a_3\right\}.$$

We will identify T' with the maximal split torus of $SL(3)$ by means of the map

$$\begin{pmatrix}A&0\\0&B\end{pmatrix}\in T'\longmapsto \mathrm{diag}(a_2,a_3,a_1^{-1}),$$

with the maximal split torus of $PGL(3)$ by means of the map

(A.13) $$\begin{pmatrix}A&0\\0&B\end{pmatrix}\in T'\longmapsto \mathrm{diag}(a_1a_2,a_1a_3,1),$$

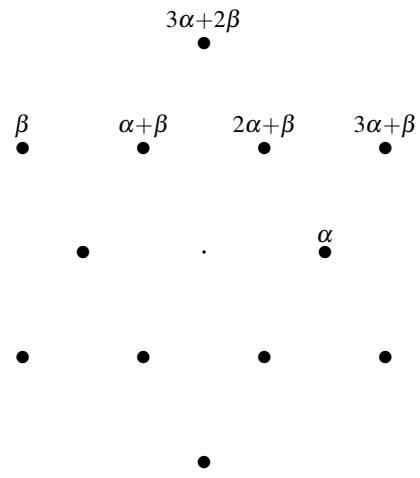
with the maximal split torus of $PGL(2)\times SL(2)$ by means of the map

(A.13′) $$\begin{pmatrix}A&0\\0&B\end{pmatrix}\in T'\longmapsto \mathrm{diag}(a_3,1)\times diag(a_1a_2,1/a_1a_2),$$

and with the maximal split torus of $(SL(2)\times SL(2))/\{\pm 1\}$ by means of the map

(A.13″) $$\begin{pmatrix}A&0\\0&B\end{pmatrix}\in T'\longmapsto \mathrm{diag}(a_1,1/a_1)\times diag(a_2,1/a_2).$$

A.4. The Chevalley basis. The embedding of $\mathfrak{g}_2$ into so(8) gives a convenient framework for doing calculations with explicit matrices. We now write down the Chevalley basis of G_2. The notation for the root system is given by the following diagram:



the root system for G_2

$$X_\alpha := \begin{pmatrix} 0 & 1 & & & & & & \\ & 0 & & & & & & \\ & & 0 & 1 & 1 & & & \\ & & & 0 & 0 & -1 & & \\ & & & & 0 & -1 & & \\ & & & & & 0 & & \\ & & & & & & 0 & -1 \\ & & & & & & & 0 \end{pmatrix}, \quad h_\alpha := \begin{pmatrix} 1 & & & & & & & \\ & -1 & & & & & & \\ & & 2 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & -2 & & \\ & & & & & & 1 & \\ & & & & & & & -1 \end{pmatrix}.$$

(Note that $X_\alpha = E_{12} + E_{34} + E_{35}$ and $\alpha \begin{pmatrix} A & \\ & B \end{pmatrix} = a_3$, where A, B are as in (A.12).)

$$X_\beta := \begin{pmatrix} 0 & & & & & & & \\ & 0 & 1 & & & & & \\ & & 0 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & -1 & \\ & & & & & & 0 & \\ & & & & & & & 0 \end{pmatrix}, \quad h_\beta := \begin{pmatrix} 0 & & & & & & & \\ & 1 & & & & & & \\ & & -1 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 1 & & \\ & & & & & & -1 & \\ & & & & & & & 0 \end{pmatrix}.$$

(Note that $X_\beta = E_{23}$ and $\beta\begin{pmatrix} A & \\ & B \end{pmatrix} = a_2 a_3^{-1}$, where A, B are as in (A.12).) For $\sigma = \alpha + \beta$:

$$X_\sigma := \begin{pmatrix} 0 & & -1 & & & & & \\ & 0 & & 1 & 1 & & & \\ & & 0 & & & & & \\ & & & 0 & & & -1 & \\ & & & & 0 & & -1 & \\ & & & & & 0 & & 1 \\ & & & & & & 0 & \\ & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 1 & & & & & & & \\ & 2 & & & & & & \\ & & -1 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 1 & & \\ & & & & & & -2 & \\ & & & & & & & -1 \end{pmatrix}.$$

(Note that $X_{\alpha+\beta} = E_{13} + E_{24} + E_{25}$.) For $\sigma = 2\alpha + \beta$:

$$X_\sigma := \begin{pmatrix} 0 & & & 1 & 1 & & & \\ & 0 & & & & 1 & & \\ & & 0 & & & & -1 & \\ & & & 0 & & & & -1 \\ & & & & 0 & & & -1 \\ & & & & & 0 & & \\ & & & & & & 0 & \\ & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 2 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & -1 & & \\ & & & & & & -1 & \\ & & & & & & & -2 \end{pmatrix}.$$

(Note that $X_{2\alpha+\beta} = E_{14} + E_{15} + E_{26}$.) For $\sigma = 3\alpha + \beta$:

$$X_\sigma := \begin{pmatrix} 0 & & & & 1 & & & \\ & 0 & & & 0 & & & \\ & & 0 & & 0 & 0 & -1 & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & & \\ & & & & & & 0 & \\ & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 1 & & & & & & & \\ & 0 & & & & & & \\ & & 1 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & -1 & & \\ & & & & & & 0 & \\ & & & & & & & -1 \end{pmatrix}.$$

(Note that $X_{3\alpha+\beta} = E_{16}$.) For $\sigma = 3\alpha + 2\beta$:

$$X_\sigma := \begin{pmatrix} 0 & & & & & 1 & & \\ & 0 & & & & 0 & -1 & \\ & & 0 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & & \\ & & & & & & 0 & \\ & & & & & & & 0 \end{pmatrix}, \quad h_\sigma := \begin{pmatrix} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 0 & & & & & \\ & & & 0 & & & & \\ & & & & 0 & & & \\ & & & & & 0 & & \\ & & & & & & -1 & \\ & & & & & & & -1 \end{pmatrix}.$$

(Note that $X_{3\alpha+2\beta} = E_{17}$.)

There is a canonical embedding $G_2 \hookrightarrow SO(8)$, given by the natural faithful representation associated to the Cayley algebra. In fact, each one–parameter unipotent subgroup U_σ associated to a short root σ of G_2 is contained in a three dimensional unipotent subgroup of SO(8) generated by three (uniquely determined) roots of D_4 which are permuted by S_3. The possibilities for this embedding

of unipotents can be tabulated as follows:

$$\begin{aligned} &U_\alpha \subset G_2 \longleftrightarrow U := \exp[E_{12}F + E_{34}F + E_{35}F], \\ &U_{2\alpha+\beta} \subset G_2 \longleftrightarrow U := \exp[E_{14}F + E_{15}F + E_{26}F], \\ &U_{\alpha+\beta} \subset G_2 \longleftrightarrow U := \exp[E_{13}F + E_{24}F + E_{25}F]. \end{aligned} \tag{A.14}$$

For the long roots of G_2, we have

$$\begin{aligned} &U_\beta \subset G_2 \longleftrightarrow U := \exp[E_{23}F], \\ &U_{3\alpha+2\beta} \subset G_2 \longleftrightarrow U := \exp[E_{17}F], \\ &U_{3\alpha+\beta} \subset G_2 \longleftrightarrow U := \exp[E_{16}F]. \end{aligned} \tag{A.15}$$

In other words, under the action of S_3 on the roots of D_4, the six stable roots correspond to the long roots of G_2 and the other 18, permuted in orbits of 3, correspond to the short roots of G_2.

A.5. Endoscopic groups of G_2. The calculation of the cuspidal endoscopic groups of G_2 is as follows. Motivated by facts about root systems of centralizers of semi-simple elements [Spal, Ch III, §12], let $\Delta := \{\alpha, \beta\}$, $\Delta_1 := \{\alpha, 3\alpha+2\beta\}$, $\Delta_2 := \{\beta, 3\alpha+\beta\}$, $\Delta^\vee := \{\alpha^\vee, \beta^\vee\}$, $\Delta_1^\vee := \{\alpha^\vee, (3\alpha+2\beta)^\vee\}$, and $\Delta_2^\vee := \{\beta^\vee, (3\alpha+\beta)^\vee\}$, where $\sigma^\vee = 2\sigma/(\sigma,\sigma)$. Representing the root α by the vector $(1,0)$ and β by $(-\frac{3}{2}, \frac{\sqrt{3}}{2})$, we find that the root lattice Q is given by

$$Q = \mathbb{Z}[\Delta] = \mathbb{Z}[(1,0), (-\frac{3}{2}, \frac{\sqrt{3}}{2})].$$

We have $\alpha^\vee = (2,0)$ and $\beta^\vee = (-1, \frac{1}{\sqrt{3}})$. Since G_2 is adjoint, its character lattice X is equal to its root lattice: $X = Q$. G_2 is also simply connected, so the weight lattice

$$\begin{aligned} P &= \{x \in \mathbb{R}^2 \mid (x, \Sigma^\vee) \subset \mathbb{Z}\} \\ &= \{(a,b) \mid (a,b)\cdot(2,0) \in \mathbb{Z},\ (a,b)\cdot(-1, \frac{1}{\sqrt{3}}) \in \mathbb{Z}\} \\ &= \mathbb{Z}[(1/2, 0), (0, \frac{\sqrt{3}}{2})] \end{aligned}$$

is also equal to the character lattice. Similarly, $Q^\vee = \mathbb{Z}[(2,0), (-1, \frac{1}{\sqrt{3}})] = P^\vee$. We have $Q_1 = \mathbb{Z}[(1,0), (0, \sqrt{3})]$, $P_1 = \mathbb{Z}[(1/2, 0), (0, \sqrt{3}/2)]$, $Q_1^\vee = \mathbb{Z}[(2,0), (0, 2/\sqrt{3})]$, and $P_1^\vee = \mathbb{Z}[(1,0), (0, 1/\sqrt{3})]$. Fix a maximal split torus ${}^L T^o$ of ${}^L G_2^o$ shared by ${}^L H^o$ and identify X with the lattice of characters of ${}^L T^o$. Take ${}^L H^o$ to be of type $A_1 \times A_1$, with root system generated by $\beta^\vee$ (which is now short) and $(3\alpha+2\beta)^\vee$ (which is not equal to $3\beta^\vee + 2\alpha^\vee$). The character lattice is equal to $X_1^\vee = X^\vee = Q^\vee$ but its root lattice $Q_1^\vee$ is strictly contained in $Q^\vee$ and $Q^\vee = X_1^\vee$ strictly contains $P^\vee$. Therefore, ${}^L H^0$ is neither simply connected nor adjoint if ${}^L H^o$ is of type $A_1 \times A_1$. This forces either $H = PGL(2) \times SL(2)$ or $H = (SL(2) \times SL(2))/\{\pm 1\}$. Following a suggestion of T. Hales, we now show that the former possibility may be discarded. The center $Z_{{}^L H^o}$ is of order two: let $z \in Z_{{}^L H^o}$ and note that $\alpha^\vee(z) = \zeta^\vee(z) = 1$ (the dual roots acting on the maximal split torus of the dual group); these imply that $\beta^\vee(z) = \pm 1$, which means that

$$Z_{{}^L H^o}(\mathbb{C}) = \{diag(\varepsilon, \varepsilon, 1, 1, 1, 1, \varepsilon, \varepsilon) \mid \text{ where } \varepsilon = \pm 1\}.$$

The image of this in $PGL(2) \times SL(2)$, under the embedding in (A.13) is trivial, so $H = (SL(2) \times SL(2))/\{\pm 1\}$. Similarly, using Δ_2 in place of Δ_1, one finds that ${}^L H^o \subset {}^L G_2^o$ is simply connected if

${}^LH^o$ is of type A_2. (This fact can also be found in [SpSt, ch II, 5.3-5.5].) Therefore $H = PGL(3)$ and $H = (SL(2) \times SL(2))/\{\pm 1\}$, as desired.

A.6. Tori of G_2. The stable conjugacy classes of Cartans in a quasi-split group G are in 1-1 correspondence with the cohomology classes in $H^1(\Gamma, W)$, where W denotes the absolute Weyl group of a fixed Cartan T of G defined over F [Ro, §4, Lemma 3]. One may also determine the Cartans from the Γ-module structure on their lattice of characters [Sat].

Let T be a Cartan of G_2 defined over F which splits over a Galois extension K. The Galois group $\Gamma_K := Gal(K/F)$ acts on the character lattice $X = X(T)$, as does the Weyl group W. Recall that T is determined (up to F-isomorphism) from the Γ_K-module X. The Γ_K-invariant submodules

$$X^{\Gamma_K} := \{\chi \in X \mid \chi^\sigma = \chi,\ \forall \sigma \in \Gamma_K\},$$

and

$$X_{\Gamma_K} := \{\chi \in X \mid \sum_{\sigma \in \Gamma_K} \chi^\sigma = 0\},$$

satisfy the property that X/X^{Γ_K} and X/X_{Γ_K} are torsion free. Let T_0 denote the annihilator of X^{Γ_K} in T and let A denote the annihilator of X_{Γ_K} in T. Then $T_0 \cap A$ is finite, $T = T_0 A$, A is the largest F-split subtorus of T, and T_0 is the largest F-anisotropic subtorus of T. Let $\Sigma_{\Gamma_K} \subset \Sigma$ denote the closed subsystem defined by $\Sigma_{\Gamma_K} := X_{\Gamma_K} \cap \Sigma$, let $\pi : X \to X/X_{\Gamma_K} \cong X(A)$, and let $\overline{\Sigma} := \pi(\Sigma - \Sigma_{\Gamma_K})$ denote the set of restricted roots relative to A. By Tate-Nakayama [L′, p. 702], the character κ of $\mathfrak{D}(T/F) \subset H^1(\Gamma_K, T)$ may be regarded as a character of the "cocycle sublattice" $X^\vee_{\Gamma_K}$, trivial on the "coboundary sublattice",

$$\partial X^\vee := \{\chi^\vee \in X^\vee \mid \chi^\vee = \sum_{\sigma \in \Gamma_K} \mu_\sigma^\sigma - \mu_\sigma,\ \text{for some } \mu_\sigma \in X^\vee\}.$$

Can the action of Γ on T be realized as an inner automorphism (defined over $\overline{F}$)? By a theorem of Chevalley, any automorphism $\sigma \in \Gamma$ may be factored into an inner automorphism and an outer automorphism of G_2 contained in $Aut(X, \Delta)$, where Δ denotes the set of simple positive roots of G_2 with respect to $T/\overline{F}$. Since G_2 has no outer automorphisms, it follows that the answer to our question is yes.

By the previous paragraph, the action of Γ on T must factor through the absolute Weyl group W. The only Cartan which had a κ-stable germ which did not vanish identically when restricted to T was the following case:
Suppose that the non-trivial action of Γ on T is realized by (a) simple reflections $< s_\alpha, s_\zeta >$, (b) the cyclic group of order 6, $< s_\beta s_\alpha >$, or (c) the cyclic group of order 3, $< (s_\beta s_\alpha)^2 >$. In the first case, T splits over a biquadratic extension E/F and $Gal(E/F)$ acts on the character lattice by $< s_\alpha, s_\zeta >$. Moreover, $T = T_G$ is isomorphic over F to $T_H = (T_1 \times T_2)/\{\pm 1\} \hookrightarrow (SL(2,F) \times SL(2,F))/\{\pm 1\}$. In this case,

$$H^1(\Gamma, T) = (H^1(\Gamma, T_1) \times H^1(\Gamma, T_2))/\{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z},$$

so the character κ must be quadratic.

In the second case, note $s_\beta s_\alpha$ acts by a counterclockwise rotation on the lattice $X^\vee_\Gamma$:

$$s_\beta s_\alpha(m\alpha^\vee + n\beta^\vee) = (2m - n)\alpha^\vee + (m - n)\beta^\vee.$$

Since

$$\sum_{i=0}^{5} (s_\beta s_\alpha)^i (m\alpha^\vee + n\beta^\vee) = 0,$$

we have $X_\Gamma^\vee = X^\vee$ (this also remains true if we replace $s_\beta s_\alpha$ by $(s_\beta s_\alpha)^2$). Since

$$s_\beta s_\alpha(-2\alpha^\vee - \beta^\vee) - (-2\alpha^\vee - \beta^\vee) = \alpha^\vee,$$

$$s_\beta s_\alpha(-3\alpha^\vee - 2\beta^\vee) - (-3\alpha^\vee - 2\beta^\vee) = \beta^\vee,$$

we have $\partial X^\vee = X^\vee$. Therefore, $H^1(\Gamma, T)$ is trivial by Tate-Nakayama duality.

In the cubic case,

$$s_\beta s_\alpha(m\alpha^\vee + n\beta^\vee) = (-3n)\alpha^\vee + (m-3n)\beta^\vee.$$

Therefore, in this case,

$$H^1(\Gamma, T) \cong \mathbb{Z}/3\mathbb{Z},$$

and κ must be a character of order 3.

All other non-split Cartans of G_2 do not contribute to the support of the 2-regular and 3-regular κ-stable germs of G_2.

A.7. The stability of the 2-regular and 3-regular unipotent orbits for G_2. I thank M. Assem for explaining to me the technique of this section.

Using the appendices above, we find that

$$\mathfrak{m} := cent_{\mathfrak{g}_2}(h_{\alpha+\beta}) = \mathfrak{t} + F \cdot X_{3\alpha+\beta} + F \cdot X_{-3\alpha-\beta},$$

where $\mathfrak{t}$ denotes the Cartan of $\mathfrak{g}_2$. Let $M := Cent_{G_2}(h_{\alpha+\beta})$ and let

$$\mathfrak{g}_2(i) := \{X \in \mathfrak{g}_2 \mid [h_{\alpha+\beta}, X] = iX\}.$$

We compute

$$\mathfrak{g}_2(2) = F \cdot X_{\alpha+\beta}.$$

The map $m \longmapsto Ad(m)X_{\alpha+\beta}$ defines a submersion $M \to \mathfrak{g}_2(2)$ [SpSt, ch III, §4.14]. By a lemma of Rao [R], the $G(F)$-orbit of $X_{\alpha+\beta}$ is $Ad(K(V(X_{\alpha+\beta}) + \mathfrak{n}(2)))$, where

$$V(X_{\alpha+\beta}) = \{ad(X)X_{\alpha+\beta} \mid X \in \mathfrak{m}\},$$

K is a maximal compact subgroup of $G_2(F)$, and

$$\mathfrak{n}(2) := \sum_{i>2} \mathfrak{g}_2(i).$$

The fact that $dim\ \mathfrak{g}_2(2) = 1$ gives us that orbit of M in $\mathfrak{g}_2(2)$ is onto, so by Rao's result the 2-regular unipotent orbit is stable.

A similar calculation shows that the 3-regular orbit is stable. (In fact, replacing $h_{\alpha+\beta}$ and $X_{\alpha+\beta}$ by h_β and X_β, we compute

$$\mathfrak{g}_2(2) = F \cdot X_\beta,$$

and the identical argument holds.)

A Lie triple for the subregular orbit is

$$X_+ = X_\beta + X_{2\alpha+\beta}, X_- = X_{-\beta} + X_{-2\alpha-\beta}, H = h_\beta + h_{2\alpha+\beta},$$

and

$$\mathfrak{g}_2(2) = F \cdot X_{\alpha+\beta} + F \cdot X_{2\alpha+\beta} + F \cdot X_{3\alpha+\beta} + F \cdot X_\beta.$$

Direct matrix calculations show that the subregular orbit is not stable. M. Assem has shown that there are 7 F-classes in the stable class, $p \neq 2, 3$.

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